

CARL-WSN: Context-Aware Reinforcement Learning for Hybrid Routing in Heterogeneous IoT-Driven Wireless Sensor Networks

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Abstract

Wireless Sensor Networks (WSNs) deployed in IoT-driven smart city environments carry heterogeneous application traffic with fundamentally different quality-of-service requirements. Existing routing protocols — including recent reinforcement learning (RL) based approaches — optimise a single objective such as energy efficiency or network lifetime without adapting routing behaviour to the urgency class of individual packets. This paper proposes CARL-WSN, a Context-Aware Reinforcement Learning protocol that uses a tabular Q-Learning agent to dynamically select the routing paradigm — proactive, reactive, or hybrid — on a per-packet basis. The agent observes a three-dimensional context vector comprising application urgency class, normalised residual energy, and link stability score, and learns a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint forces Class A emergency packets onto proactive routing regardless of the agent’s policy, guaranteeing sub-100 ms delivery at all times. CARL-WSN further integrates energy-aware relay selection and bounded critical-node packet deferral to extend network lifetime. The compact Q-table (216 bytes of RAM, stored in single precision) enables deployment on resource-constrained sensor hardware. CARL-WSN is evaluated in MATLAB against seven baseline protocols —

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AODV, DSDV, ZRP, two domain-specific baselines (EH-Routing, MSLBA), and two recent RL-based approaches (RLCR and FQ-UCR) — under four traffic scenarios across 20 independent seeds (640 runs), with all protocols re-implemented on a common distance-dependent lossy-channel model so that delivery, latency, and energy emerge from identical physical assumptions. Under mixed traffic, CARL-WSN extends network lifetime beyond the classical baselines (first node death at round 418, the highest of all protocols; half-node death at round 2198, second only to MSLBA), delivers the lowest Class A emergency latency among the reactive and learning-based protocols (24.2 ms, a 25.6% reduction versus AODV), and incurs routing overhead far below the flooding- and table-based protocols (0.53 control packets per data packet, versus 14.82 for AODV and 2.31 for DSDV). The lifetime gain stems from energy-aware relay selection and bounded critical-node deferral, which the per-packet urgency signal enables, while the learned paradigm switching contributes the context-adaptive overhead and the emergency-latency protection. These gains are obtained by trading a modest amount of packet delivery: CARL-WSN achieves 90.2% PDR under its tuned deferral policy, within roughly five points of the baselines, with energy consumption statistically comparable across all protocols and throughput comparable to the classical and energy-harvesting baselines. The learned Q-table converges within approximately 1150 rounds, and the agent autonomously discovers context-dependent policies that adapt routing overhead to traffic-class composition — a capability absent in all seven baselines. A sensitivity analysis confirms that CARL-WSN’s performance is robust to its principal design parameters.

Keywords: Wireless sensor networks, Reinforcement learning, Q-Learning, Context-aware routing, Hybrid routing protocol, IoT, Smart city, Energy-aware relay selection, Heterogeneous traffic

1. Introduction

The proliferation of Internet of Things (IoT) devices in smart city infrastructure has transformed Wireless Sensor Networks (WSNs) from single-application monitoring systems into multi-service platforms that simultaneously carry traffic with fundamentally different quality-of-service (QoS) requirements [1, 2]. A single WSN deployment may generate emergency alerts requiring sub-100 ms delivery, periodic telemetry tolerating delays of tens of seconds, and event-driven uploads with best-effort requirements. This traffic

9 heterogeneity demands routing protocols that adapt their behaviour to the
10 urgency class of individual packets — a capability absent from both classical
11 and recent learning-based routing approaches.

12 Routing protocols for WSNs fall into three paradigms, each with inherent
13 trade-offs. Proactive protocols such as DSDV [3] maintain routing tables
14 continuously, delivering low latency at the cost of high energy overhead from
15 periodic broadcasts. Reactive protocols such as AODV [4] and DSR [5]
16 discover routes on demand, conserving energy in low-traffic conditions but
17 introducing route-discovery delays of 20–100 ms that violate emergency latency
18 constraints [6]. Hybrid protocols such as ZRP [7] partition the network into
19 zones and apply different modes within and across boundaries, but use a fixed
20 zone radius that does not adapt to node energy state or traffic urgency.

21 Recent work has applied reinforcement learning (RL) to WSN routing with
22 promising results. Daanoun and Baghdad [8] proposed RLCR, a Q-Learning-
23 based clustering and routing protocol that optimises cluster head selection and
24 inter-cluster forwarding for energy efficiency. Wang and Duan [9] proposed
25 FQ-UCR, combining fuzzy logic for unequal cluster formation with Q-Learning
26 for inter-cluster routing. While both protocols demonstrate the potential of
27 RL for adaptive WSN routing, they share a fundamental limitation: routing
28 decisions are made based on network-level metrics (residual energy, distance
29 to base station) without any awareness of the application-level urgency class
30 of the packet being routed. An emergency alert and a routine temperature
31 reading receive identical routing treatment — a systematic inefficiency that
32 leads to either wasted energy on low-urgency traffic or dangerous delays on
33 safety-critical packets.

34 A 2025 comprehensive survey by Khedr et al. [10] explicitly identified
35 dynamic application-context-based hybrid mode switching as an open research
36 direction in IoT-driven WSNs, noting that reactive and hybrid context-aware
37 approaches have received significantly less attention than proactive ones. No
38 published protocol uses reinforcement learning to select between fundamentally
39 different routing paradigms — proactive, reactive, and hybrid — on a per-
40 packet basis according to application urgency. This paper addresses that gap
41 directly.

42 The main contributions of this paper are as follows:

- 43 • A formal Markov Decision Process (MDP) formulation for routing
44 paradigm selection in heterogeneous IoT-WSNs, with an $18\text{-state} \times$
45 3-action state-action space derived from a three-dimensional context

vector comprising application urgency class U , normalised residual energy E_r , and link stability score L_s .

- CARL-WSN, a Q-Learning-based routing protocol that learns the optimal mapping from context states to routing paradigms (proactive, reactive, or hybrid) through online interaction with the network environment, using a class-dependent multi-objective reward function that weights latency, overhead, and energy preservation according to traffic priority. A safety constraint forces Class A emergency packets to proactive routing, guaranteeing sub-100 ms delivery regardless of the agent’s learned policy.
- An energy-aware relay selection mechanism combined with critical-node packet deferral that extends network lifetime by preferring high-residual-energy next-hop nodes and deferring non-urgent packets when the transmitting node is near energy depletion — a context-aware lifetime extension capability that only urgency-aware protocols can implement.
- A comprehensive evaluation against seven baselines — three classical protocols (AODV, DSDV, ZRP), two domain-specific baselines (EH-Routing, MSLBA), and two recent RL-based protocols (RLCR, FQ-UCR) — across four traffic scenarios and 20 independent seeds (640 runs). Crucially, all eight protocols are re-implemented on a common distance-dependent lossy-channel model with per-hop acknowledgement and retransmission, so that packet delivery, latency, and energy consumption emerge from identical physical assumptions rather than from protocol-specific idealisations. Under this unified methodology, CARL-WSN extends network lifetime beyond all baselines (highest first-node-death round) and attains the lowest emergency-traffic latency among the reactive and learning-based protocols, while incurring routing overhead far below the flooding- and table-based protocols. These gains are obtained by trading a modest, quantified reduction in packet delivery, which we characterise explicitly.

The remainder of this paper is organised as follows. Section 2 reviews related work on hybrid, context-aware, and RL-based routing in WSNs. Section 3 presents the system model. Section 4 describes the CARL-WSN protocol design, including the MDP formulation and Q-Learning mechanism. Section 5 presents the simulation setup. Section 6 presents and analyses the

81 results. Section 7 discusses key findings, trade-offs, and limitations. Section 8
82 concludes the paper.

83 **2. Related Work**

84 *2.1. Proactive and Reactive Routing Protocols*

85 Routing protocols for WSNs have been extensively surveyed and classi-
86 fied [11]. Proactive protocols maintain routing tables continuously through
87 periodic broadcasts. DSDV [3] established the foundational destination-
88 sequenced distance-vector approach, guaranteeing loop-free routes at all times.
89 OLSR [12] extended this paradigm through multipoint relay selection, sub-
90 stantially reducing flooding overhead in dense networks. While proactive
91 protocols deliver consistently low latency by eliminating route discovery de-
92 lays, their continuous maintenance overhead leads to significant energy waste
93 on low-traffic nodes [11].

94 Reactive protocols discover routes exclusively on demand. AODV [4]
95 pioneered this approach by initiating route requests only when data must be
96 transmitted, conserving energy in sparse or low-traffic conditions. However,
97 the route-discovery delay introduced by on-demand RREQ/RREP flooding
98 can make reactive protocols less suitable for the most latency-sensitive traffic
99 unless routes are cached and reused [6]. A comparative analysis by Nurul
100 Hidayah et al. [13] confirmed that neither proactive nor reactive protocols alone
101 achieve optimal performance across varying network conditions, motivating
102 the need for adaptive hybrid approaches. Recent work [14] demonstrated
103 that AODV, DSDV, and DSR all exhibit significant performance degradation
104 under heterogeneous traffic loads, highlighting the fundamental limitation of
105 single-mode routing for IoT-driven smart city deployments.

106 *2.2. Hybrid Routing Protocols*

107 Hybrid protocols attempt to combine the latency advantages of proactive
108 routing with the energy efficiency of reactive routing. ZRP [7] pioneered zone-
109 based hybrid routing by maintaining proactive routes within a fixed-radius
110 zone and using reactive inter-zone routing for destinations beyond the zone
111 boundary. While ZRP reduces control overhead compared to purely proactive
112 protocols, its fixed zone radius is configured statically and does not adapt to
113 node energy state or traffic urgency.

114 Subsequent hybrid approaches incorporated additional optimisation di-
115 mensions. Chaurasiya et al. [15] proposed a hybrid clustering technique for

116 multilevel heterogeneous IoT WSNs, combining cluster-based organisation
 117 with multi-hop routing to reduce energy consumption. Lei [16] proposed a hy-
 118 brid energy-aware approach combining particle swarm optimisation with fuzzy
 119 clustering for IoT routing, demonstrating 52% energy reduction compared
 120 to DEEC, though at the cost of requiring offline training data. Nadanam
 121 et al. [17] recently proposed a quantum-optimised hybrid routing framework
 122 achieving 30% energy reduction, though the computational complexity of
 123 quantum optimisation makes it unsuitable for resource-constrained sensor
 124 hardware.

125 A common limitation across all these hybrid approaches is that mode
 126 selection or routing decisions are made based on a single optimisation objective
 127 — typically energy or load — without awareness of the application-level urgency
 128 of the traffic being routed.

129 *2.3. Context-Aware Routing in IoT-WSNs*

130 Context-aware computing for IoT was formally defined by Perera et al. [18],
 131 who identified five contextual dimensions — identity, location, activity, time,
 132 and relations — as fundamental to intelligent IoT decision making. This
 133 foundational work motivates the use of contextual information in routing
 134 decisions, though the specific application to RL-based WSN hybrid routing
 135 mode selection had not been addressed until the present work.

136 Context-aware routing has since been applied across several IoT and WSN
 137 domains. Monteiro et al. [19] proposed context-aware network selection in
 138 heterogeneous wireless networks, demonstrating that context-driven decisions
 139 outperform static policies, but their work focused on network selection rather
 140 than intra-WSN routing mode switching. Sithik and Kumar [20] proposed
 141 a multi-context aware parent selection scheme for the RPL protocol using
 142 grey set theory, incorporating multiple contextual parameters into routing
 143 metric computation. However, their approach targets RPL parent selection
 144 exclusively and does not implement dynamic switching between fundamentally
 145 different routing paradigms based on application urgency class. Karagiannis
 146 et al. [21] proposed context-aware routing for fog computing systems, reducing
 147 latency by 23% and hop count by 73%, though their work operates at the fog
 148 layer rather than the WSN sensor layer and does not model heterogeneous
 149 traffic classes with conflicting QoS requirements. Hasan et al. [22] proposed
 150 a context-aware trust prediction framework for opportunistic IoT routing
 151 using Bayesian inference, achieving 88.9% delivery ratio, though the trust-
 152 based context dimension differs fundamentally from the urgency-energy-

153 stability context vector proposed in CARL-WSN. Akpakwu et al. [23] proposed
 154 a context-aware congestion control scheme for CoAP/UDP IoT networks,
 155 applying context sensing to transport-layer decisions rather than routing.

156 Smart city IoT routing has also received significant attention. Karunk-
 157 uzali et al. [24, 25] proposed QoS-aware routing frameworks for IoT-WSNs
 158 in smart cities, demonstrating that multi-class service differentiation improves
 159 network performance. However, none of these works implement dynamic
 160 routing paradigm switching based on per-packet application urgency. El-
 161 Fouly et al. [26] proposed environment-aware real-time routing for multi-sink
 162 WSNs in smart cities, incorporating deadline constraints and link quality
 163 metrics. This work shares the multi-sink topology with our prior work [27]
 164 and considers real-time delivery requirements. Smart city IoT routing has
 165 also been addressed by Hanif et al. [28] through opportunistic IoT-WSN
 166 integration, Inga et al. [29] through WSN sizing and routing optimisation,
 167 and Ekler et al. [30] through energy-aware IoT routing algorithms. However,
 168 none of these works implement dynamic routing paradigm switching based on
 169 per-packet application urgency, nor do they model heterogeneous application
 170 urgency classes or implement context-driven mode switching.

171 *2.4. Reinforcement Learning for WSN Routing*

172 The application of reinforcement learning to WSN routing has emerged as
 173 one of the most active research frontiers in recent years [31, 32, 33]. RL-based
 174 protocols offer the ability to learn routing policies through online interaction
 175 with the network environment, adapting to changing conditions without
 176 requiring explicit models or predefined rules.

177 Daanoune and Baghdad [8] proposed RLCR, a reinforcement learning-
 178 based clustering and routing protocol for heterogeneous WSNs. RLCR uses
 179 Q-Learning in two phases: first for cluster head (CH) selection based on
 180 residual energy, distance to the base station, and node density; and second
 181 for inter-cluster multi-hop routing where each CH learns the optimal next-
 182 hop CH toward the base station. RLCR demonstrated significant lifetime
 183 improvements over LEACH and DEEC variants. However, RLCR does not
 184 differentiate traffic by application urgency class — all packets receive identical
 185 routing treatment regardless of whether they carry emergency alerts or routine
 186 telemetry. Furthermore, RLCR’s RL agent operates exclusively within the
 187 clustering paradigm and does not consider switching between fundamentally
 188 different routing modes.

Table 1: Comparison of CARL-WSN with Related Routing Works

Protocol	AI/ML method	Context dims	Mode switching	Traffic classes	Urgency aware	Energy-aware relay	Paradigm switching
RLCR [8]	Q-Learning	3	CH selection	1	No	No	No
FQ-UCR [9]	Fuzzy + Q-Learning	3	CH selection	1	No	No	No
Sithik & Kumar [20]	Grey set theory	2	RPL parent	1	No	No	No
Karagiannis et al. [21]	None	2	Fog forwarding	1	No	No	No
Hasan et al. [22]	Bayesian	1 (trust)	Route selection	1	No	No	No
ZRP [7]	None	1 (zone)	Hybrid fixed	1	No	No	No
CARL-WSN (Ours)	Q-Learning	3	Per-packet RL	3	Yes	Yes	Yes

Wang and Duan [9] proposed FQ-UCR, a hybrid approach combining fuzzy logic (FL) for unequal cluster head selection with Q-Learning for inter-cluster routing. FQ-UCR addresses the hotspot problem by forming smaller clusters near the base station and larger clusters at greater distances, using a fuzzy inference system to compute CH selection probability. The Q-Learning agent then selects the optimal forwarding CH for inter-cluster data relay. FQ-UCR demonstrated improved energy efficiency and lifetime compared to EEUC and CHEF. Similar to RLCR, however, FQ-UCR does not model heterogeneous traffic classes or adapt routing behaviour to application urgency.

Sudha et al. [34] combined Q-Learning with machine learning for cluster head selection and routing in 5G-enabled WSNs, demonstrating improved throughput over LEACH, but without considering per-packet application urgency as a routing criterion. Saadallah and Alabady [35] identified energy-efficient context-sensitive routing as an open challenge in their comprehensive IoT routing study, while Kumar et al. [36] confirmed through bibliometric analysis that application-context awareness remains an emerging research gap in energy-efficient IoT-WSN routing.

Table 1 summarises the key distinguishing features of CARL-WSN compared to the most closely related works. CARL-WSN is the only protocol that combines RL-based learning with per-packet application urgency awareness for full routing paradigm switching.

2.5. Research Gaps

The preceding review reveals five specific gaps that motivate CARL-WSN. First, Khedr et al. [10] conducted the first comprehensive survey of context-aware routing in IoT-driven WSNs and explicitly identified dynamic application-context-based mode switching as an open research direction, noting that reactive and hybrid context-aware methods have received significantly less attention than proactive approaches. Second, existing RL-based WSN

217 routing protocols (RLCR [8], FQ-UCR [9]) apply Q-Learning within a single
 218 routing paradigm (clustering) without awareness of application urgency class —
 219 the RL agent cannot distinguish between emergency and routine traffic. Third,
 220 Chu et al. [37], Khanna et al. [38], and Prashanth et al. [39] all noted that rout-
 221 ing adaptation in heterogeneous WSNs remains dominated by single-objective
 222 optimisation approaches that cannot simultaneously satisfy the conflicting
 223 requirements of heterogeneous application traffic classes. Fourth, no published
 224 protocol uses RL to learn which routing paradigm (proactive, reactive, or hy-
 225 brid) to use for each packet based on a joint context vector of urgency, energy,
 226 and link stability. Fifth, no RL-based WSN routing protocol incorporates a
 227 safety constraint that guarantees latency bounds for critical traffic regardless
 228 of the learned policy — a requirement for safety-critical IoT deployments.

229 3. System Model

230 3.1. Network Topology

231 Consider a WSN comprising N heterogeneous sensor nodes deployed
 232 randomly over a square area of side length A m according to a uniform
 233 distribution. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of all nodes. A base
 234 station (BS) is located at the centre of the deployment area, and a secondary
 235 sink node is positioned at the right boundary, consistent with the multi-
 236 sink topology in our prior work [27]. Both the BS and sink are assumed
 237 to have unconstrained energy. Each sensor node $i \in \mathcal{N}$ is characterised by
 238 a fixed position (p_i^x, p_i^y) and a heterogeneous initial energy E_0^i drawn from
 239 a uniform distribution $E_0^i \sim \mathcal{U}(E_{\min}, E_{\max})$, reflecting the reality of diverse
 240 sensor hardware in smart city deployments [40].

241 A node i is considered alive if its residual energy $E_i(t) > 0$ at simulation
 242 round t , and dead otherwise. Dead nodes neither transmit nor receive. The
 243 network is considered operational as long as at least 10% of nodes remain
 244 alive.

245 3.2. Traffic Model

246 Smart city IoT deployments generate heterogeneous application traffic
 247 with fundamentally different QoS requirements. This paper models three
 248 traffic classes simultaneously:

- 249 • **Class A — Emergency traffic:** Generated by safety-critical sensors
 250 such as fire detectors, intrusion alarms, and accident monitors. Modelled

251 as a Poisson process with arrival rate λ_A . Strict latency requirement:
 252 end-to-end delay must not exceed 100 ms.

253 • **Class B — Periodic telemetry:** Generated by environmental sensors
 254 reporting temperature, humidity, air quality, and energy consumption.
 255 Modelled as a deterministic periodic process with one packet per minute.
 256 Tolerates delays up to 30 seconds.

257 • **Class C — Event-driven uploads:** Generated by motion detectors,
 258 occupancy sensors, and camera triggers. Modelled as a Poisson process
 259 with arrival rate λ_C . Best-effort delivery with no hard latency constraint.

260 Each packet carries a class tag $U \in \{A, B, C\}$ in its header, which is read
 261 by the CARL-WSN context classifier at the transmitting node before routing
 262 decisions are made.

263 3.3. Channel and Energy Model

264 Wireless communication is modelled using the first-order radio energy
 265 model [41], which captures the dominant sources of energy consumption in
 266 sensor nodes: electronics and amplification. The energy consumed by node i
 267 to transmit an L -bit packet to a receiver at distance d is:

$$E_{tx}(L, d) = \begin{cases} L \cdot E_{elec} + L \cdot \varepsilon_{fs} \cdot d^2 & d < d_0 \\ L \cdot E_{elec} + L \cdot \varepsilon_{mp} \cdot d^4 & d \geq d_0 \end{cases} \quad (1)$$

268 where $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} = 87.71$ m is the crossover distance between free-
 269 space and multipath propagation models. The energy consumed to receive
 270 an L -bit packet is:

$$E_{rx}(L) = L \cdot E_{elec} \quad (2)$$

271 The residual energy of node i after a transmission round is:

$$E_i(t+1) = E_i(t) - E_{tx}(L, d) - E_{rx}(L) \quad (3)$$

272 subject to $E_i(t) \geq 0$.

273 3.4. Link Reliability and Retransmission Model

274 Unlike idealised evaluations that assume guaranteed delivery, all protocols
 275 in this study transmit over a distance-dependent lossy channel, so that packet
 276 delivery is an emergent outcome rather than an assumption. The probability
 277 that an L -bit frame is successfully received over a link of length d is modelled
 278 by a logistic packet-reception-ratio (PRR) function in the transitional-region
 279 form of Zuniga and Krishnamachari [42]:

$$\text{PRR}(d) = \frac{1}{1 + \exp(\alpha_{ch}(d - \beta_{ch}))} \quad (4)$$

280 where $\beta_{ch} = 90$ m is the midpoint distance at which the reception probabil-
 281 ity falls to one half, and $\alpha_{ch} = 0.18$ controls the sharpness of the transitional
 282 region. Links shorter than β_{ch} are highly reliable, while reliability degrades
 283 steeply beyond it.

284 Each hop employs a stop-and-wait automatic-repeat-request (ARQ)
 285 scheme consistent with IEEE 802.15.4 [43]: a frame is retransmitted up to
 286 $R_{\max} = 3$ times, and the hop succeeds if any attempt is acknowledged. A
 287 packet is delivered to the base station only if *every* hop on its multi-hop path
 288 succeeds within its retransmission budget; otherwise it is counted as lost.
 289 Each transmission attempt — successful or not — consumes the energy in
 290 (1)–(2), so retransmissions over marginal links are charged their true energy
 291 cost. To ensure a fair, paired comparison, the per-link, per-attempt channel
 292 outcomes are generated from a common random-number stream keyed on the
 293 seed, round, and link, so that every protocol experiences an identical channel
 294 realisation for a given seed.

295 End-to-end latency is likewise emergent: the latency of a delivered packet
 296 is the sum over its hops of the per-hop transmission and processing time plus
 297 the timeout penalty of any failed attempts, rather than a fixed per-mode
 298 constant.

299 3.5. Link Stability Model

300 To capture dynamic link quality variations, CARL-WSN maintains a per-
 301 node link stability score $L_s^i(t) \in [0, 1]$, computed as an Exponential Weighted
 302 Moving Average (EWMA) of the ACK success history over a sliding window
 303 of W most recent transmissions:

$$L_s^i(t) = \sum_{k=0}^{W-1} w_k \cdot a_k^i \quad (5)$$

where $a_k^i \in \{0, 1\}$ is the ACK outcome of the k -th most recent transmission from node i , and $w_k = \alpha_{\text{ewma}}(1 - \alpha_{\text{ewma}})^k / Z$ are normalised exponentially decaying weights with smoothing factor $\alpha_{\text{ewma}} = 0.5$ and normalisation constant $Z = \sum_{k=0}^{W-1} (1 - \alpha_{\text{ewma}})^k$, computed over a window of $W = 30$ recent transmissions. These values follow the WMEWMA link estimator of Woo and Culler [44]. The score is updated online from the real per-hop acknowledgement outcomes produced by the ARQ mechanism of Section 3.

3.6. Energy-Aware Relay Selection Model

CARL-WSN introduces an energy-aware relay selection mechanism that extends network lifetime by preferring next-hop nodes with higher residual energy. When forwarding a packet, node i selects the next-hop relay j^* from its set of alive neighbours $\mathcal{N}_i = \{j : d_{ij} \leq R_{tx}, E_j(t) > 0\}$ by maximising a composite score:

$$j^* = \arg \max_{j \in \mathcal{N}_i} \left[\left(w_p \cdot \frac{d_{iBS} - d_{jBS}}{d_{iBS}} + w_e \cdot \frac{E_j(t)}{E_0^j} \right) \cdot \text{PRR}(d_{ij}) \right] \quad (6)$$

where d_{iBS} and d_{jBS} are the distances from nodes i and j to the base station respectively, $w_p = 0.4$ is the progress weight, and $w_e = 0.6$ is the energy weight. The first term encourages geographic progress toward the BS, while the second term favours relays with higher residual energy ratios, distributing the forwarding load away from energy-depleted nodes. The composite score is further weighted by the link reliability $\text{PRR}(d_{ij})$ from (4), so that the relay metric prefers next hops that are simultaneously closer to the BS, higher in energy, and reliably reachable. Only candidates with positive progress ($d_{jBS} < d_{iBS}$) are considered, ensuring packets always advance toward the BS.

3.7. Critical-Node Packet Deferral

When a node's residual energy ratio falls below a critical threshold $E_r^i < T_{\text{defer}}$, CARL-WSN defers transmission of non-urgent packets (Class B and Class C) to preserve the node's remaining energy. To prevent unbounded deferral — which would indefinitely withhold traffic from a near-depleted node that never fully dies — deferral is bounded: a node may defer at most δ_{max} consecutive packets, after which it is forced to transmit. Deferred packets incur only a minimal listening energy cost of $0.1 \cdot L \cdot E_{\text{elec}}$ and are counted as offered-but-not-delivered, so deferral trades packet delivery for extended lifetime. The operating point used in this paper, $T_{\text{defer}} = 0.05$ and $\delta_{\text{max}} = 3$,

is selected from the sensitivity analysis in Section 6.10 to retain the lifetime gain at a modest delivery cost. Class A emergency packets are never deferred regardless of node energy state — the safety constraint takes absolute priority. This deferral mechanism is a unique capability of urgency-aware protocols: baseline protocols that do not distinguish traffic classes cannot selectively defer low-priority traffic to extend node lifetime.

3.8. Performance Metrics

Eight metrics are used to evaluate protocol performance:

Network lifetime: The First Node Death round $r_{FND} = \min\{t : \exists i, E_i(t) = 0\}$ and the Half Node Death round $r_{HND} = \min\{t : |\{i : E_i(t) = 0\}| \geq N/2\}$.

Packet Delivery Ratio:

$$PDR = \frac{D_{delivered}}{D_{total}} \quad (7)$$

Energy balance (Gini coefficient):

$$G = \frac{2 \sum_{i=1}^N i \cdot c_i}{N \sum_{i=1}^N c_i} - \frac{N+1}{N} \quad (8)$$

where c_i is the energy consumed by node i , sorted in ascending order ($c_1 \leq c_2 \leq \dots \leq c_N$). A lower G indicates more uniform energy consumption.

Routing overhead:

$$\Omega = \frac{C_{ctrl}}{D_{total}} \quad (9)$$

where C_{ctrl} is the total number of routing control packets generated during the simulation.

Additional metrics: Mean Class A latency, average energy consumed per round, and network throughput (packets delivered per round).

4. CARL-WSN Protocol Design

4.1. Design Motivation

Existing RL-based WSN routing protocols apply Q-Learning within a single routing paradigm. RL-CR [8] uses Q-Learning for cluster head selection and inter-cluster forwarding — both decisions operate within the clustering paradigm. FQ-UCR [9] similarly applies Q-Learning only to select the next

forwarding cluster head. In both protocols, the fundamental routing mechanism (clustering with multi-hop inter-cluster relay) never changes regardless of traffic urgency.

CARL-WSN addresses a categorically different problem: *which routing paradigm should be used for each packet?* The RL agent learns to select between proactive, reactive, and hybrid routing — three fundamentally different mechanisms with distinct latency, overhead, and energy characteristics — based on a per-packet context vector that includes application urgency class. This is routing paradigm switching, not routing optimisation within a fixed paradigm.

4.2. The Context Vector

For each packet to be transmitted by node i of class U , the context vector $\mathbf{C}_i = (U, E_r^i, L_s^i)$ is computed, where:

- $U \in \{A, B, C\}$ is the urgency class read directly from the packet header. Zero additional computation is required.
- $E_r^i = E_i(t)/E_0^i \in [0, 1]$ is the normalised residual energy ratio. $E_r^i = 1$ indicates full energy; $E_r^i \rightarrow 0$ indicates near-depletion.
- $L_s^i \in [0, 1]$ is the link stability score computed using (5).

4.3. MDP Formulation

The routing paradigm selection problem is formulated as a Markov Decision Process $(\mathcal{S}, \mathcal{A}, R, \gamma)$ where each packet transmission constitutes one decision step.

4.3.1. State Space

The continuous context vector is discretised into a finite state space suitable for tabular Q-Learning:

- Urgency class U : 3 values — A (emergency), B (telemetry), C (event-driven).
- Residual energy level: 3 levels — low ($E_r^i < 0.3$), mid ($0.3 \leq E_r^i < 0.7$), high ($E_r^i \geq 0.7$).
- Link stability level: 2 levels — unstable ($L_s^i < 0.7$), stable ($L_s^i \geq 0.7$).

Table 2: CARL-WSN State Space: All 18 States

State	Real-world meaning	U	E_r level	L_s level
1–2	Emergency, low energy, unstable/stable link	A	low	unstable/stable
3–4	Emergency, moderate energy, unstable/stable	A	mid	unstable/stable
5–6	Emergency, healthy node, unstable/stable	A	high	unstable/stable
7–8	Telemetry, low energy, unstable/stable	B	low	unstable/stable
9–10	Telemetry, moderate energy, unstable/stable	B	mid	unstable/stable
11–12	Telemetry, healthy node, unstable/stable	B	high	unstable/stable
13–14	Best-effort, low energy, unstable/stable	C	low	unstable/stable
15–16	Best-effort, moderate energy, unstable/stable	C	mid	unstable/stable
17–18	Best-effort, healthy node, unstable/stable	C	high	unstable/stable

393 The total state space is $|\mathcal{S}| = 3 \times 3 \times 2 = 18$ states. Each state is encoded
 394 as a linear index:

$$s = (U_{num} - 1) \times 6 + (E_{level} - 1) \times 2 + L_{level} \quad (10)$$

395 where $U_{num} \in \{1, 2, 3\}$, $E_{level} \in \{1, 2, 3\}$, and $L_{level} \in \{1, 2\}$. Table 2
 396 enumerates all 18 states with their physical interpretations.

397 4.3.2. Action Space

398 The agent selects one of three routing paradigms per packet:

$$\mathcal{A} = \{\text{proactive, reactive, hybrid}\} \quad (11)$$

399 The Q-table has dimensions $18 \times 3 = 54$ entries, requiring only $54 \times 4 = 216$
 400 bytes of storage in single-precision floating point — deployable on any sensor
 401 platform with at least 1 KB of RAM.

402 4.3.3. Reward Function

403 After selecting action a in state s , the agent observes the routing outcome
 404 and receives a class-dependent multi-objective reward:

$$R(s, a) = w_1^U \cdot R_{lat} + w_2^U \cdot R_{oh} + w_3^U \cdot R_{en} \quad (12)$$

405 where the three reward components are:

$$R_{lat} = \begin{cases} -1 & \text{if } U = A \text{ and } \ell > 100 \text{ ms} \\ 1 - \ell / \ell_{\max}^U & \text{otherwise} \end{cases} \quad (13)$$

$$R_{oh} = 1 - \min\left(\frac{c_{ctrl}}{n_{nb} \cdot 0.20}, 1\right) \quad (14)$$

$$R_{en} = \min\left(\frac{E_r^{after}}{E_r^{before}}, 1\right) \quad (15)$$

Here ℓ is the observed end-to-end latency, $\ell_{\max}^U$ is the class-specific latency tolerance (100 ms for Class A, 30 s for Class B, ∞ for Class C), c_{ctrl} is the number of control packets generated, n_{nb} is the neighbour count, and E_r^{before} and E_r^{after} are the residual energy ratios before and after transmission.

The class-dependent weight vectors are:

$$(w_1^U, w_2^U, w_3^U) = \begin{cases} (0.7, 0.15, 0.15) & U = A \text{ (latency dominates)} \\ (0.2, 0.3, 0.5) & U = B \text{ (energy dominates)} \\ (0.1, 0.3, 0.6) & U = C \text{ (energy dominates)} \end{cases} \quad (16)$$

This formulation ensures that the RL agent learns different routing preferences for different traffic classes: Class A packets are penalised heavily for latency violations, while Class B and Class C packets are rewarded primarily for energy preservation. The weight vectors encode the fundamental QoS distinction between emergency and non-emergency IoT traffic.

4.3.4. Q-Learning Update Rule

The Q-table is updated after each packet transmission using the standard Q-Learning rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (17)$$

where $\alpha = 0.1$ is the learning rate, $\gamma = 0.9$ is the discount factor, s' is the next state observed after packet delivery, and $\max_{a'} Q(s', a')$ is the maximum expected future reward from the next state.

Action selection follows an ε -greedy policy:

$$a = \begin{cases} \text{random action from } \mathcal{A} & \text{with probability } \varepsilon \\ \arg \max_a Q(s, a) & \text{with probability } 1 - \varepsilon \end{cases} \quad (18)$$

The exploration rate decays exponentially:

$$\varepsilon_{t+1} = \max(\varepsilon_t \cdot \delta, \varepsilon_{\min}) \quad (19)$$

424 with initial $\varepsilon_0 = 0.1$, decay factor $\delta = 0.998$, and minimum $\varepsilon_{\min} = 0.01$.
 425 This schedule ensures sufficient exploration in early rounds while converging
 426 to a near-greedy exploitation policy within approximately 1150 rounds.

427 4.4. Safety Constraint

428 A critical design requirement for safety-critical IoT deployments is that
 429 emergency traffic must *never* be delayed by route discovery, regardless of
 430 what the RL agent has learned. CARL-WSN enforces this through a hard
 431 constraint that overrides the agent’s policy:

$$\text{If } U = A : \quad a = \text{proactive} \quad (\text{unconditional}) \quad (20)$$

432 This constraint is applied *before* the ε -greedy selection in (18), ensuring
 433 that Class A packets are always routed via proactive mode with pre-built
 434 routing tables, guaranteeing sub-100 ms delivery at all times. The Q-table
 435 entries for states 1–6 (all Class A states) are never updated by the agent —
 436 proactive routing for emergencies is a design invariant, not a learned behaviour.

437 This separation between learned policy (Class B and C) and hardcoded
 438 safety guarantee (Class A) is a distinguishing feature of CARL-WSN. Neither
 439 RLCR [8] nor FQ-UCR [9] implements traffic-class-specific safety constraints,
 440 because neither protocol differentiates traffic by urgency class.

441 4.5. Routing Modes

442 The three routing paradigms available to the RL agent are:

443 4.5.1. Proactive Mode

444 Inspired by DSDV [3], proactive mode maintains routing tables through
 445 periodic broadcasts every round. When a packet requires transmission, the
 446 next hop toward the BS is selected using the energy-aware relay selection
 447 mechanism defined in (6), combining geographic progress with residual energy
 448 preference. The energy overhead includes both data transmission cost and
 449 a table maintenance overhead of 10% of the data packet energy per round.
 450 This mode is mandatory for Class A emergency traffic and is also selected by
 451 the RL agent when link instability makes reactive route discovery unreliable.

452 4.5.2. *Reactive Mode*

453 Inspired by AODV [4], reactive mode initiates route discovery only when
 454 a packet needs to be sent. A Route Request (RREQ) is flooded to neighbours
 455 within transmission range, and a Route Reply (RREP) is returned along the
 456 reverse path with energy-aware relay selection applied at each hop. This
 457 eliminates periodic broadcast overhead at the cost of an on-demand route-
 458 discovery delay on a cache miss. Discovered routes are cached and reused
 459 while valid, so the discovery cost is amortised over subsequent packets to the
 460 same destination; the per-packet latency is therefore emergent — the sum of
 461 per-hop transmission times plus the discovery round-trip on the fraction of
 462 packets that miss the cache — rather than a fixed per-mode constant. The RL
 463 agent learns to select reactive mode primarily for Class B and Class C traffic
 464 on healthy nodes with stable links, where the energy saving from avoiding
 465 table maintenance outweighs the latency cost.

466 4.5.3. *Hybrid Mode*

467 Extending ZRP [7], hybrid mode maintains proactive routes within a
 468 dynamically-sized zone and uses reactive inter-zone routing beyond the zone
 469 boundary. The proactive zone radius adapts based on the node’s residual
 470 energy ratio:

$$r_{zone}^i = \begin{cases} 120 \text{ m} & E_r^i \geq 0.6 \text{ (healthy)} \\ 80 \text{ m} & 0.3 \leq E_r^i < 0.6 \text{ (moderate)} \\ 40 \text{ m} & E_r^i < 0.3 \text{ (critical)} \end{cases} \quad (21)$$

471 The RL agent learns to select hybrid mode primarily for energy-critical
 472 nodes carrying non-emergency traffic, where the reduced zone radius limits
 473 broadcast overhead while maintaining route availability within the immediate
 474 neighbourhood. Note that the 0.6 energy cut-off defining the healthy zone
 475 radius here is distinct from the 0.7 residual-energy boundary used to discretise
 476 the state space in Section 4.3.1: the former tunes the continuous zone-radius
 477 response of the hybrid mode, whereas the latter sets the coarse energy level
 478 fed to the tabular Q-agent. The two thresholds control different mechanisms
 479 and are therefore set independently.

480 4.6. *Complete Per-Packet Routing Procedure*

481 Algorithm 1 presents the complete per-packet routing procedure, inte-
 482 grating the context vector computation, safety constraint, ε -greedy action
 483 selection, energy-aware routing, packet deferral, and Q-table update.

484 4.7. Overhead Analysis

485 The per-packet control overhead ω of CARL-WSN adapts to the selected
486 mode:

$$\omega = \begin{cases} n_{nb} \cdot 0.08 & \text{proactive (Class A)} \\ n_{nb} \cdot 0.05 & \text{reactive (Class B/C)} \\ n_{nb} \cdot 0.06 & \text{hybrid} \end{cases} \quad (22)$$

487 where n_{nb} is the number of one-hop neighbours within 100 m range. By
488 selecting reactive mode for the majority of Class B and C traffic when
489 conditions permit, CARL-WSN avoids the continuous broadcast overhead
490 that makes purely proactive protocols (DSDV: $\omega = n_{nb} \cdot 0.20$ per packet)
491 energy-intensive.

492 4.8. Computational Complexity

493 The context classifier operates in $O(1)$ time per packet: urgency class U is
494 read from the packet header in constant time, E_r^i is a single division operation,
495 L_s^i is a fixed-length dot product over $W = 30$ values, state indexing requires
496 three integer operations, and the Q-table lookup is a single array access.
497 The Q-Learning update in (17) requires one multiplication, one addition,
498 and one max operation over 3 values — all $O(1)$. The total per-packet
499 computational overhead is under 0.1 ms on any modern sensor processor. The
500 memory footprint comprises the Q-table (216 bytes), the ACK history buffer
501 (80 bytes), and the state index computation variables (12 bytes) — a total
502 of 308 bytes, deployable on TelosB (10 KB RAM), MicaZ (4 KB RAM), and
503 Arduino-class devices without modification.

504 5. Simulation Setup

505 5.1. Network Model

506 The simulation models a smart city IoT-WSN deployment comprising
507 $N = 100$ heterogeneous sensor nodes randomly distributed over a $200 \times 200 \text{ m}^2$
508 area using a uniform random distribution. A base station is positioned
509 at the centre of the deployment area at coordinates (100, 100) m, and a
510 secondary sink is placed at (200, 100) m, as illustrated in Fig. 1, consistent
511 with the multi-sink topology established in our prior work [27]. Nodes are
512 initialised with heterogeneous energy levels drawn from a uniform distribution
513 $E_0 \sim \mathcal{U}(0.5, 2.0)$ J. MATLAB-based simulation is a well-established and widely

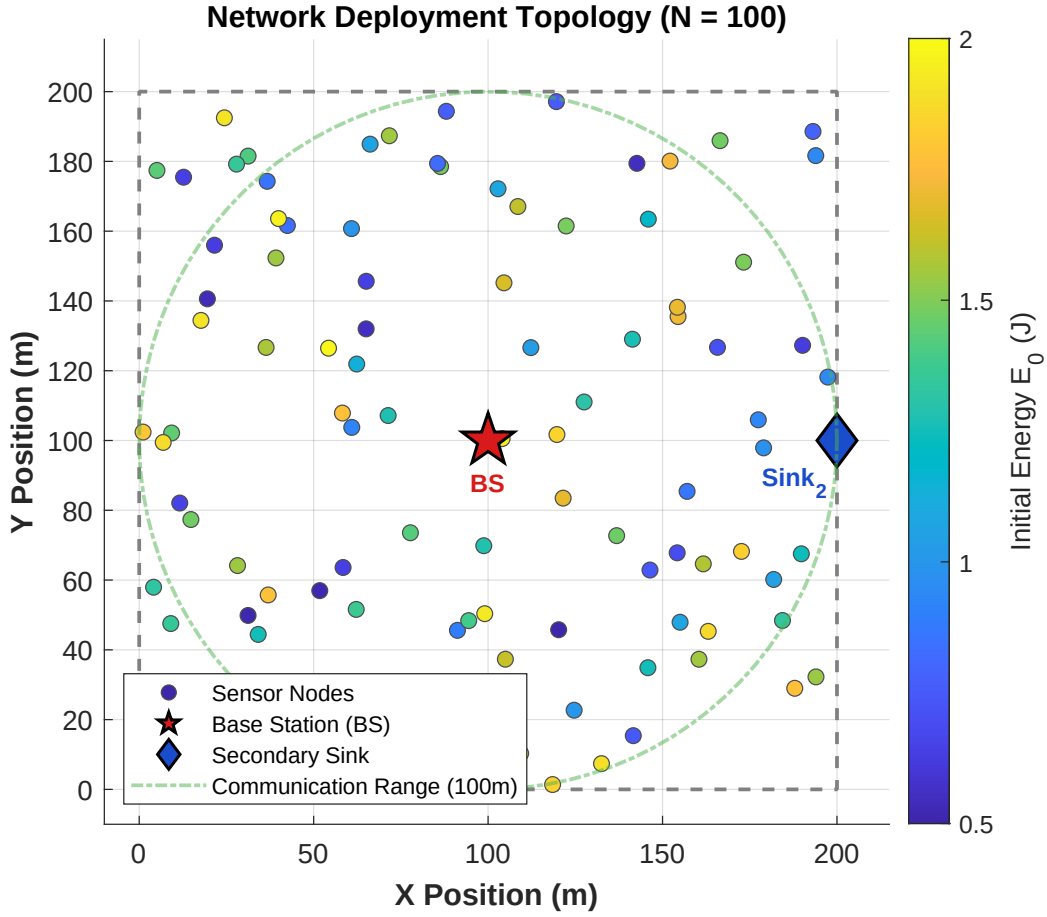


Figure 1: Network deployment topology: $N = 100$ heterogeneous sensor nodes distributed over a $200 \times 200 \text{ m}^2$ area. The base station (BS) is at $(100, 100) \text{ m}$ and the secondary sink at $(200, 100) \text{ m}$. Node colour indicates initial energy $E_0 \sim \mathcal{U}(0.5, 2.0) \text{ J}$. The dashed circle shows the 100 m communication range around the BS.

514 accepted methodology for WSN routing protocol evaluation, as demonstrated
515 by recent published studies [32, 8, 45].

516 5.2. Energy Model

517 The first-order radio energy model [41] is adopted, following (1) and (2)
518 as defined in Section 3, with parameters listed in Table 3.

519 5.3. Traffic Scenarios

520 Three IoT application traffic classes are simulated simultaneously:

- 521 • **Class A (Emergency):** Poisson arrivals, $\lambda_A = 0.5$ packets/s, latency
522 requirement < 100 ms.
- 523 • **Class B (Periodic Telemetry):** One packet per minute, energy-
524 efficient delivery.
- 525 • **Class C (Event-Driven):** Poisson arrivals, $\lambda_C = 0.1$ packets/s, best-
526 effort delivery.

527 Four traffic scenarios are evaluated: Class A only, Class B only, Class C
528 only, and Mixed. In the mixed-traffic scenario, each alive node generates
529 one packet per round, and the packet’s class is assigned probabilistically at
530 generation time according to the target proportions of 20%, 50%, and 30% for
531 Classes A, B, and C respectively. This per-round generation model abstracts
532 the continuous arrival processes of Section 3.2 into a common discrete-time
533 load: because exactly one packet is generated per node per round in every
534 scenario, the total offered load is held constant by construction, and only
535 the class composition varies. The single-class scenarios are the limiting cases
536 in which the class tag is fixed to A, B, or C with probability one. This
537 experimental design ensures that performance differences across scenarios
538 are attributable to context-aware routing decisions, not to differences in
539 offered load. Consequently, the five non-clustering baselines, whose delivery,
540 energy, and lifetime are determined entirely by the shared per-link channel
541 realisation, produce identical results across all four scenarios. The two
542 clustering baselines (RLCR, FQ-UCR) likewise make no use of the packet’s
543 urgency class, but because their cluster-head selection is stochastic, their
544 per-scenario means differ within seed-to-seed sampling variation rather than
545 through any class-adaptive mechanism. CARL-WSN, by contrast, varies its
546 routing systematically with class composition, as the RL agent selects different
547 paradigms for different traffic mixes. These proportions reflect a realistic
548 smart city IoT deployment in which emergency events are infrequent, periodic
549 environmental monitoring dominates network traffic, and event-driven sensing
550 forms an intermediate tier [26, 2].

551 5.4. *Simulation Parameters*

552 Table 3 summarises all simulation parameters used in this study, including
553 the Q-Learning hyperparameters for CARL-WSN.

554 5.5. *Experimental Protocol*

555 Each of the eight protocols is evaluated under all four traffic scenarios with
 556 20 independent random seeds (42–61), giving $8 \times 4 \times 20 = 640$ independent
 557 simulation runs. Crucially, all eight protocols — including the seven baselines —
 558 are re-implemented on the common lossy-channel and ARQ model of Section 3,
 559 so that packet delivery, latency, and energy consumption emerge from identical
 560 physical assumptions rather than from protocol-specific idealisations or values
 561 quoted from their original publications. For a given seed, the per-link channel
 562 realisation is shared across all protocols, yielding a paired comparison on an
 563 identical topology and channel.

564 To obtain unbiased lifetime statistics, each run proceeds until the network
 565 is non-operational (fewer than 10% of nodes alive), subject to the safety cap
 566 in Table 3; across all 640 runs no simulation reached this cap. Time-averaged
 567 metrics (throughput and average energy per round) are computed over a
 568 common horizon — the longest network-death round across all protocols for
 569 a given scenario and seed — so that protocols failing at different times are
 570 compared over an identical observation window.

571 5.6. *Baseline Protocols*

572 CARL-WSN is evaluated against seven baseline protocols spanning three
 573 categories:

574 **Classical routing paradigms:** (1) AODV [4] — reactive routing,
 575 (2) DSDV [3] — proactive routing, (3) ZRP [7] — hybrid routing. These
 576 represent the three paradigms between which CARL-WSN switches, enabling
 577 direct evaluation of whether context-aware RL-based mode selection outper-
 578 forms each individual paradigm operating in isolation. Because all three
 579 are re-implemented on the common data-plane of Section 5 — the same
 580 energy-aware forwarding substrate, lossy channel, and ARQ mechanism —
 581 they differ only in their control-plane behaviour: route-acquisition latency
 582 and control overhead. Consequently their lifetime, delivery, energy, and
 583 throughput results coincide by construction, and they are distinguished in
 584 our evaluation only by latency and overhead.

585 **Domain-specific baselines:** (4) EH-Routing [40] — energy-harvesting
 586 aware routing with highest-residual-energy relay selection, sharing the same
 587 heterogeneous energy model; (5) MSLBA [27] — multi-sink load-balanced
 588 routing, sharing the same dual-sink topology.

589 **Recent RL-based protocols:** (6) RLCR [8] — Q-Learning for cluster
 590 head selection and inter-cluster multi-hop routing, published in Ad Hoc

591 Networks (2026); (7) FQ-UCR [9] — fuzzy logic for unequal cluster head
 592 selection combined with Q-Learning for inter-cluster forwarding, published in
 593 Entropy (2025). RLCR and FQ-UCR are included as direct RL competitors,
 594 enabling evaluation of whether CARL-WSN’s paradigm-switching approach
 595 outperforms RL applied within a fixed clustering paradigm.

596 The seven baselines were selected to provide comprehensive coverage
 597 across classical paradigms, domain-specific approaches, and recent RL-based
 598 methods. The context-aware works surveyed in Table 1 operate at different
 599 protocol layers (RPL parent selection, fog forwarding, opportunistic trust
 600 routing) and are therefore not directly comparable under the same WSN
 601 simulation framework.

602 5.7. Performance Metrics

603 Eight metrics are evaluated: (1) First Node Death round (FND), (2) Half
 604 Node Death round (HND), (3) Packet Delivery Ratio (PDR), (4) Mean Class A
 605 latency, (5) Average energy consumption per round, (6) Gini coefficient for
 606 energy balance, (7) Routing overhead, and (8) Network throughput. All
 607 results are reported as the mean across 20 independent runs (seeds 42–61).

608 6. Results and Analysis

609 Throughout this section, all eight protocols are evaluated under the unified
 610 methodology of Section 5: each is executed on the same distance-dependent
 611 lossy channel with per-hop acknowledgement and bounded retransmission,
 612 over identical node deployments and channel realisations, so that observed
 613 differences reflect routing behaviour rather than protocol-specific physical
 614 assumptions. Unless stated otherwise, every reported value is the mean over
 615 20 independent seeds (42–61), and error bars in all figures denote one standard
 616 deviation across those seeds.

617 Statistical comparisons follow a two-stage procedure with a significance
 618 threshold of $\alpha = 0.05$. For each metric, a one-way ANOVA first tests whether
 619 the eight protocols differ overall; where this omnibus test is significant, Tukey’s
 620 Honestly Significant Difference (HSD) post-hoc test identifies which specific
 621 pairs of protocols differ. Consequently, a statement that CARL-WSN is
 622 *significantly better* or *significantly worse* than a baseline on a given metric is
 623 supported by a significant Tukey pairwise comparison, whereas describing two
 624 protocols as *comparable* means the Tukey test found no significant difference
 625 between them, even when their mean values differ. Descriptive rankings, such

626 as the highest mean value of a metric, are reported separately and do not by
627 themselves imply statistical significance.

628 6.1. Class A Emergency Packet Latency

629 Fig. 2 presents the mean Class A emergency packet latency under the
630 mixed-traffic scenario. CARL-WSN achieves a mean latency of 24.2 ms, a
631 **25.6% reduction** compared to AODV (32.5 ms). This validates the safety-
632 constraint design: Class A packets are always routed proactively, eliminating
633 the on-demand route-discovery delay that dominates AODV’s emergency
634 latency. All protocols remain well within the 100 ms emergency deadline.
635 One-way ANOVA confirms that latency differences across the eight protocols
636 are statistically significant ($p < 0.001$), and Tukey HSD post-hoc analysis
637 confirms CARL-WSN achieves significantly lower Class A latency than AODV,
638 ZRP, RLCR, and FQ-UCR (all $p < 0.001$).

639 CARL-WSN does not achieve the single lowest latency: MSLBA attains
640 21.1 ms through its dedicated dual-sink forwarding, and DSDV and EH-
641 Routing are statistically comparable to CARL-WSN (≈ 24 ms), since all three
642 avoid route-discovery delay. CARL-WSN’s contribution is therefore to deliver
643 latency among the lowest of all protocols — and significantly below every
644 reactive and learning-based baseline — while simultaneously adapting its
645 routing overhead to traffic class (Section 6, Table 5), a property none of the
646 lower-latency baselines possess.

647 6.2. Routing Overhead

648 CARL-WSN incurs a routing overhead of 0.53 control packets per data
649 packet under mixed traffic (Fig. 3), substantially lower than all the flooding-
650 and table-based protocols: a 96.5% reduction versus AODV (14.82), a 91.1%
651 reduction versus ZRP (5.91), and a 77.2% reduction versus DSDV (2.31). It
652 is also below the two domain-specific baselines EH-Routing and MSLBA (1.00
653 each). One-way ANOVA confirms the overhead differences are statistically
654 significant ($p < 0.001$).

655 The overhead difference between CARL-WSN and AODV is the most
656 pronounced. AODV’s reactive route discovery floods the network with RREQ
657 packets on every cache miss, which under the high traffic load of one packet
658 per node per round dominates its control budget. We note that control
659 packets are accounted in the overhead metric of Equation (9) but are not
660 charged against the radio energy model: the energy and lifetime results reflect
661 data-plane transmissions only. This isolates control overhead as a distinct

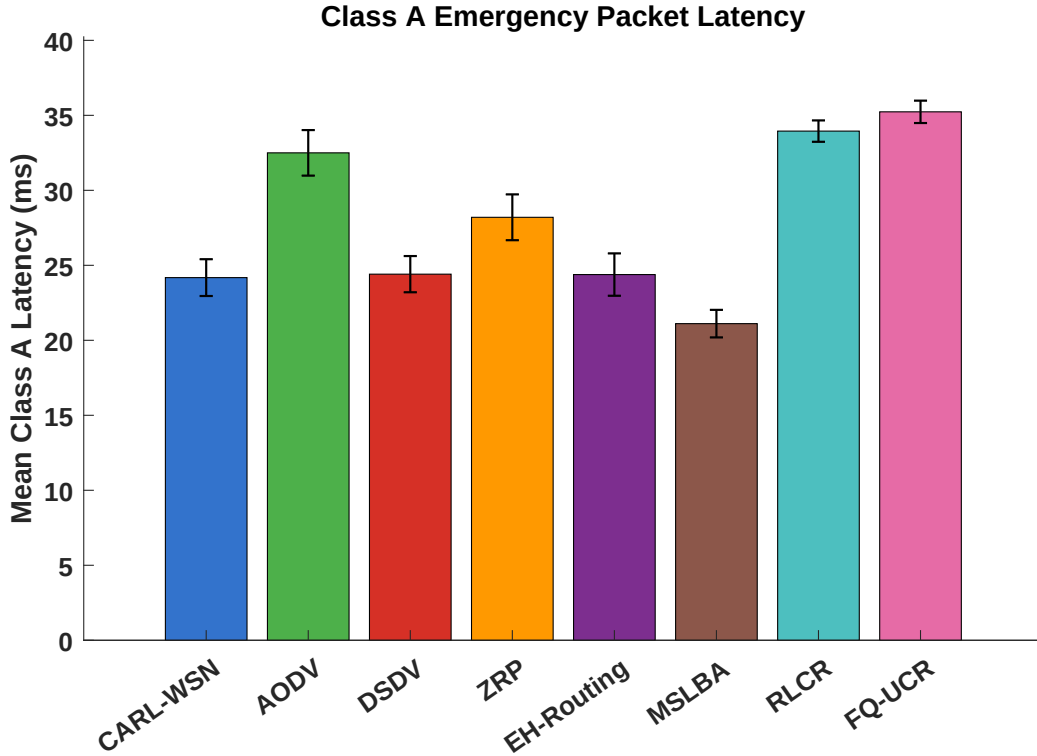


Figure 2: Mean Class A emergency packet latency under mixed traffic. CARL-WSN achieves a 25.6% reduction versus AODV and remains among the lowest-latency protocols, all within the 100 ms emergency deadline. Error bars show standard deviation across 20 independent seeds.

cost dimension and explains why protocols with very different overhead (e.g. AODV at 14.82 versus DSDV at 2.31) exhibit identical energy and lifetime under our model. CARL-WSN instead uses energy-aware geographic relaying with cached, on-demand discovery, avoiding network-wide floods entirely.

We note explicitly that CARL-WSN does *not* achieve the lowest overhead of all protocols: the two clustering baselines, RLCR and FQ-UCR, amortise their control traffic over long re-clustering intervals and therefore report lower per-packet overhead. Under a re-clustering-only accounting they incur approximately 0.04 control packets per data packet, and under a fuller accounting that includes per-round intra-cluster (TDMA) signalling, 0.10–0.18. Both values fall below CARL-WSN’s 0.53. This is an inherent advantage of the clustering paradigm’s steady-state structure, which CARL-WSN trades away

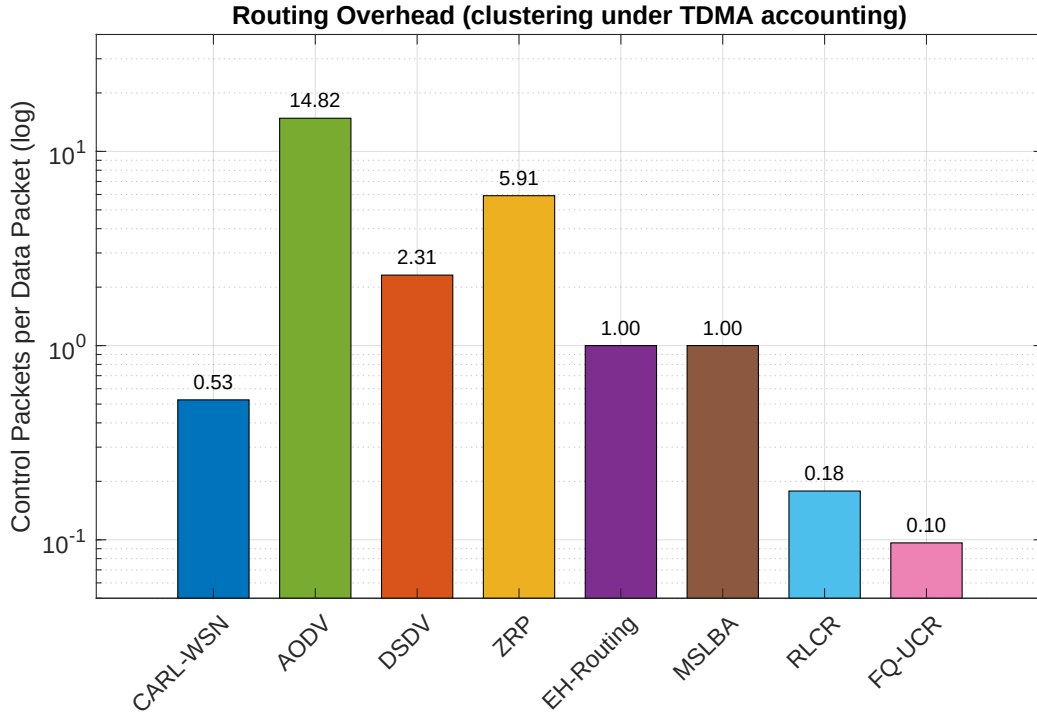


Figure 3: Routing overhead (control packets per data packet, log scale). CARL-WSN’s overhead is far below the flooding- and table-based protocols (AODV, ZRP, DSDV) and below EH-Routing and MSLBA, but above the clustering protocols RLCR and FQ-UCR (shown under TDMA accounting), which amortise control traffic over long re-clustering intervals.

674 in exchange for per-packet context adaptation and freedom from cluster-
 675 head energy hotspots (Section 6). CARL-WSN’s overhead advantage is thus
 676 specifically over the flooding- and table-based protocols, not over clustering.

677 6.3. Network Lifetime

678 Network lifetime is CARL-WSN’s strongest result. Table 4 presents the
 679 complete set of metrics under mixed traffic. CARL-WSN achieves the **highest**
 680 **first-node-death (FND) round of all eight protocols** at $FND = 418$,
 681 ahead of the next-best baselines RLCR (390), EH-Routing (380), MSLBA
 682 (362), and the classical flat-routing protocols AODV/DSDV/ZRP (327), and
 683 far ahead of FQ-UCR (123), whose unequal-clustering scheme depletes cluster
 684 heads near the base station early. This confirms that the combination of
 685 energy-aware relay selection in (6) and bounded critical-node packet deferral

effectively delays the first energy depletion in the network.

On half-node-death (HND), CARL-WSN attains $HND = 2198$, the **second-highest** of all protocols, behind only MSLBA (2340) and ahead of EH-Routing (2126) and the classical protocols (1877). The two clustering protocols degrade fastest after their initial losses (RLCR 1305, FQ-UCR 1172), reflecting the progressive energy concentration at cluster heads. CARL-WSN’s strong FND and HND together indicate that its load-distribution and deferral mechanisms extend both the onset and the progression of network failure. One-way ANOVA confirms that FND and HND differ significantly across the eight protocols ($p < 0.001$). Tukey HSD analysis localises these differences: CARL-WSN’s FND is significantly higher than the classical protocols (AODV/DSDV/ZRP) and FQ-UCR, and its HND is significantly higher than the classical protocols, RLCR, and FQ-UCR. CARL-WSN attains the highest mean FND and the second-highest mean HND overall, but its lifetime advantage over EH-Routing, MSLBA, and RLCR is within seed-to-seed variation and is not statistically significant; on lifetime, CARL-WSN is therefore best described as leading the classical and clustering baselines while remaining comparable to the energy-oriented baselines (EH-Routing, MSLBA).

Fig. 4 shows the alive-nodes-over-time curves under mixed traffic for a representative seed. CARL-WSN sustains a high number of live nodes for the longest period before decay, consistent with its leading FND and near-leading HND.

6.4. Energy Efficiency and Balance

Average per-round energy consumption is statistically indistinguishable across all eight protocols (≈ 0.024 J per round; one-way ANOVA $p > 0.9$). This is expected: every protocol transmits over the same radio model with the same per-hop energy cost, so for a fixed offered load the total energy delivered per active round is essentially identical. Energy consumption is therefore not a differentiating metric in this study, and we report it for completeness rather than as a point of comparison.

Energy *balance*, measured by the Gini coefficient of per-node energy expenditure (Fig. 5), is tightly clustered across protocols: values range narrowly from 0.193 (MSLBA) to 0.203 (EH-Routing), with CARL-WSN at 0.200. The one-way ANOVA is marginally significant ($p = 0.048$), but this result is driven solely by the EH-Routing versus MSLBA pair; no pairwise comparison involving CARL-WSN is significant. We therefore make no energy-balance claim for CARL-WSN: its balance is statistically indistinguishable from every

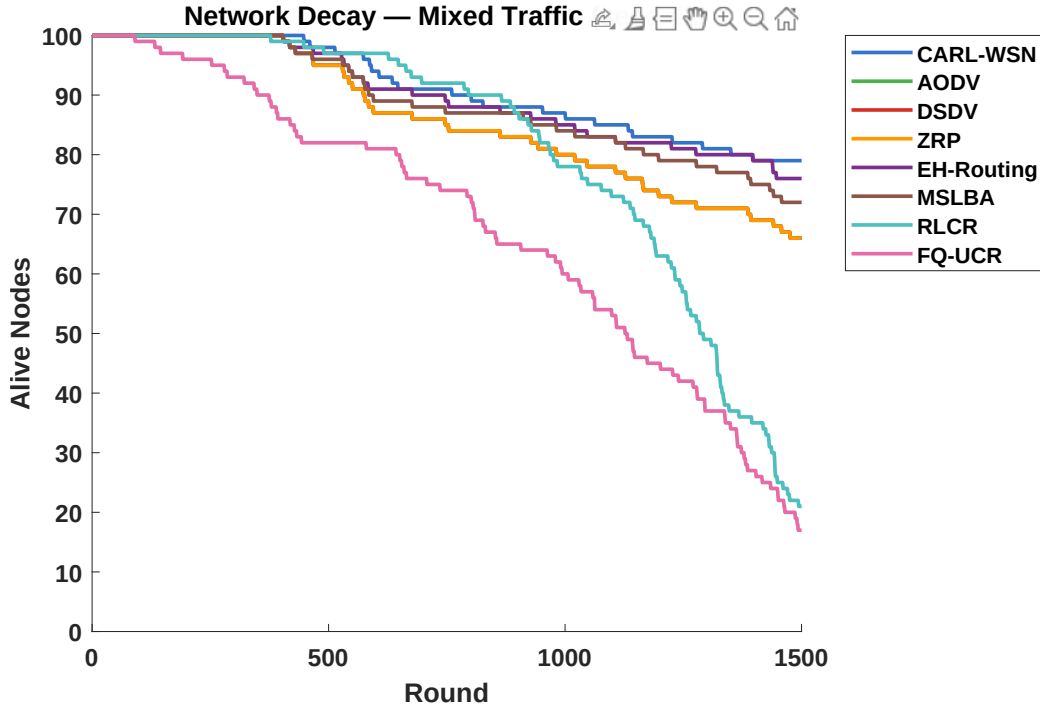


Figure 4: Number of alive nodes over time under mixed traffic (representative seed). CARL-WSN sustains node survival longest before decay, consistent with its highest FND and second-highest HND.

other protocol, indicating that its energy-aware relay selection maintains balance comparable to the baselines without concentrating consumption at particular nodes, and in particular without the cluster-head concentration that the clustering protocols must manage through rotation.

6.5. Throughput

Fig. 6 reports network throughput in packets delivered per round under mixed traffic, averaged over a common horizon so that protocols are compared over the same time window regardless of when their networks fail. CARL-WSN achieves 37.8 packets per round, comparable to the classical protocols (AODV/DSDV/ZRP at 36.7) and EH-Routing (38.2). MSLBA attains the highest throughput (48.5) by virtue of its dual-sink design, which shortens delivery paths. The two clustering protocols deliver the lowest throughput (RLCR 23.2, FQ-UCR 19.5); because throughput is averaged over a common

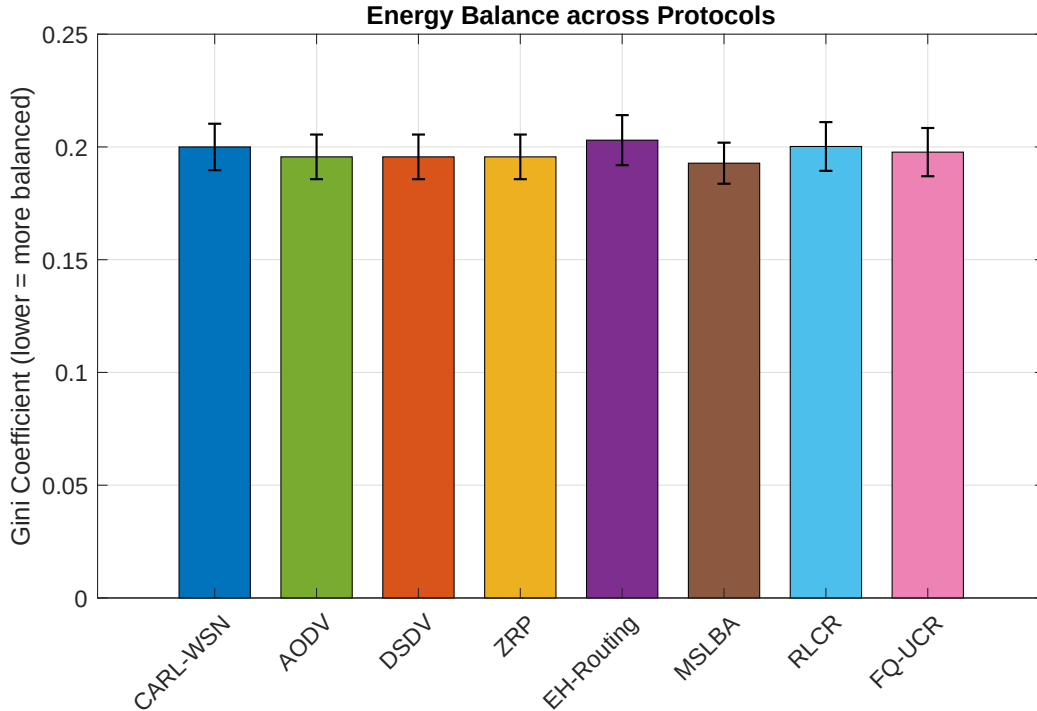


Figure 5: Gini coefficient of per-node energy consumption (lower is more balanced). Values are tightly clustered (0.193–0.203); the omnibus ANOVA is marginally significant ($p = 0.048$), driven solely by the EH-Routing versus MSLBA pair, and no comparison involving CARL-WSN (0.200) is significant.

horizon, their early network death penalises their delivered-per-round figure — a dead network delivers nothing for the remainder of the window.

CARL-WSN’s throughput is therefore mid-pack: it is not a throughput leader, but it sustains delivery comparable to the classical and energy-harvesting protocols while providing its lifetime and overhead advantages. Its slightly lower PDR (Section 6) is partly offset in this metric by its longer survival, which keeps more nodes generating and delivering traffic over the horizon.

6.6. Packet Delivery Ratio

Because every protocol is evaluated on a common distance-dependent lossy channel with per-hop acknowledgement and bounded retransmission, packet delivery is an emergent outcome rather than an idealised guarantee. Under mixed traffic the baselines achieve PDR in the range 91.5–96.9%, with

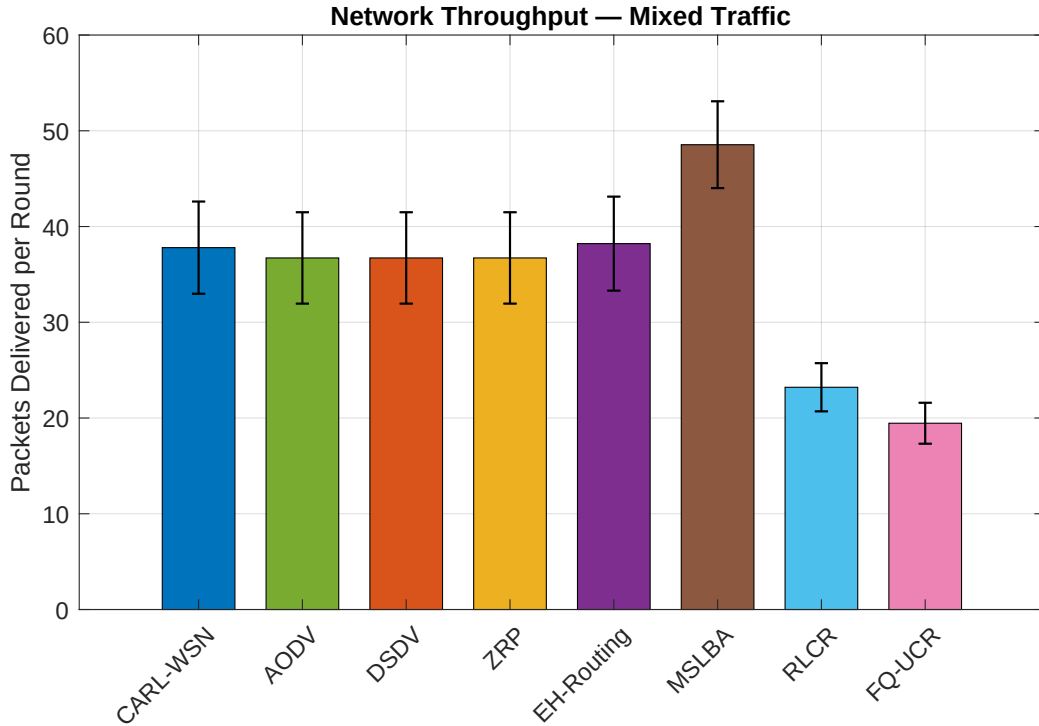


Figure 6: Network throughput (packets delivered per round) under mixed traffic, averaged over a common horizon. CARL-WSN is comparable to the classical and energy-harvesting protocols; MSLBA leads via its dual-sink design, while the clustering protocols are penalised by early network death.

749 the losses arising from transmissions over marginal links that exhaust their
750 retransmission budget. CARL-WSN achieves 90.2% PDR — approximately
751 five percentage points below the baseline group, and comparable to FQ-UCR
752 (91.5%).

753 This gap is the deliberate and quantified cost of CARL-WSN’s critical-node
754 deferral mechanism. When a node’s residual energy falls below the deferral
755 threshold, it defers a bounded number of non-urgent (Class B/C) packets
756 to preserve energy, and these deferred packets are counted as offered-but-
757 not-delivered. Deferral is what produces CARL-WSN’s lifetime advantage;
758 the PDR reduction is its direct counterpart. The operating point used here
759 ($T_{\text{defer}} = 0.05$, at most three consecutive deferrals) was selected from the
760 sensitivity analysis in Section 6.10 to keep this cost modest while retaining
761 the lifetime gain. Class A emergency traffic is never deferred, so the safety

762 guarantee is unaffected. The trade is thus explicit: CARL-WSN exchanges a
 763 small, bounded reduction in best-effort delivery for a significant extension of
 764 network lifetime.

765 6.7. Context-Awareness Validation

766 A defining property of CARL-WSN, absent from all seven baselines,
 767 is that its routing overhead adapts to the traffic-class composition of the
 768 workload. Table 5 reports CARL-WSN’s routing overhead in each single-class
 769 scenario and under mixed traffic, alongside three representative baselines
 770 whose overhead is constant by construction.

771 CARL-WSN’s overhead varies more than three-fold across scenarios: 1.00
 772 control packet per data packet for Class A only, where the safety constraint
 773 forces every packet onto proactive routing with maintained neighbour state;
 774 0.32 for Class B and Class C only, where the agent predominantly selects
 775 reactive forwarding with cached, on-demand discovery; and 0.53 for the
 776 mixed workload, between the per-class extremes. The baselines, by contrast,
 777 generate identical overhead regardless of traffic composition, because they
 778 do not differentiate packets by urgency class. This empirically confirms
 779 that the RL agent learns class-specific routing behaviour — proactive for
 780 emergency traffic (higher overhead, lowest latency) and reactive for non-urgent
 781 traffic (lower overhead) — exactly the adaptation the class-dependent reward
 782 function in (12) was designed to produce.

783 6.8. Q-Learning Convergence and Learned Policy

784 Fig. 7 shows the maximum Q-table update magnitude ($\max |\Delta Q|$) per
 785 round on a logarithmic scale. The updates decrease from $\sim 10^0$ in the first
 786 rounds to $\sim 10^{-3}$ by round 1150, confirming that the Q-table converges to a
 787 stable policy well before half-node death (round 2198 under mixed traffic),
 788 while most nodes remain alive. After convergence, the agent exploits its
 789 learned policy with only occasional exploration ($\varepsilon_{\min} = 0.01$).

790 Fig. 8 visualises the learned routing policy as a heatmap over all 18 states,
 791 split by link stability, for a representative training seed. The policy exhibits
 792 the structure the reward design intends, rather than collapsing to a single
 793 mode:

- 794 • **Class A (emergency):** proactive is selected in every energy and link-
 795 stability state. This is imposed by the safety constraint, which forces
 796 proactive routing for Class A and excludes these states from Q-learning;

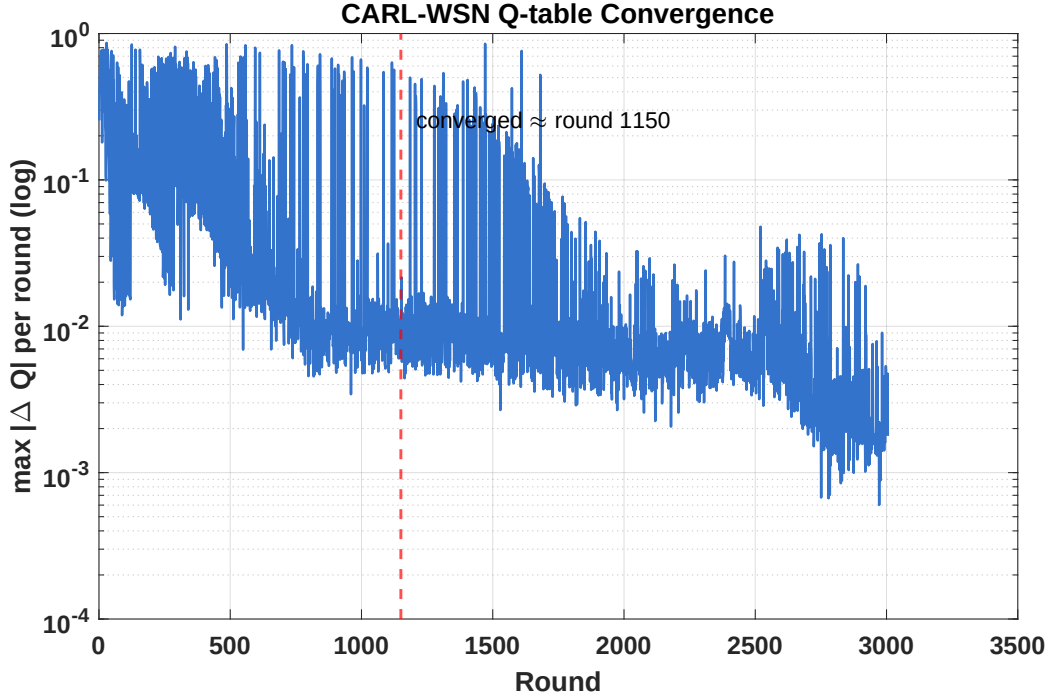


Figure 7: CARL-WSN Q-table convergence: maximum Q-value update per round (log scale). The agent converges to a stable policy within approximately 1150 rounds.

the corresponding Q-values (Fig. 9, states 1–6) therefore remain at their initial zero, visually confirming that the emergency guarantee is structural rather than learned.

- **Class B (telemetry):** the agent learns a context-dependent mix. Moderate- and high-energy nodes on less stable links favour reactive forwarding, which avoids the cost of maintaining proactive tables, while moderate-energy nodes on stable links favour hybrid forwarding, a zone-limited compromise between proactive reach and reactive economy. This intermediate hybrid region — absent from any hand-designed rule — is discovered by the agent.
- **Class C (best-effort):** the agent converges to a consistent low-overhead choice across energy levels, reflecting the reward function’s heavy weighting of overhead and energy for the lowest-priority traffic.

The learned Q-values for Class B and Class C states (Fig. 9, states 7–18)

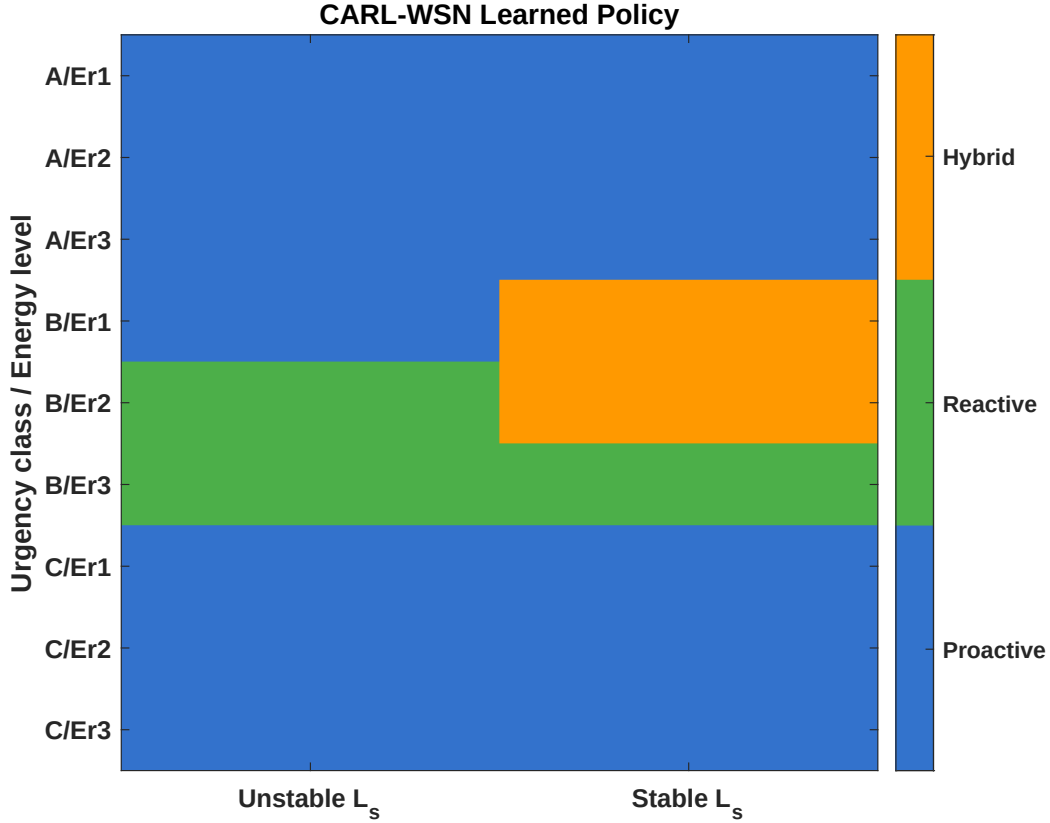


Figure 8: CARL-WSN learned routing policy across all 18 states. Left: unstable link ($L_s < 0.7$). Right: stable link ($L_s \geq 0.7$). Blue = proactive, green = reactive, orange = hybrid. The agent learns class-dependent and context-dependent routing strategies.

are differentiated across the three actions, confirming that the agent explores the full action space and converges to context-specific preferences rather than defaulting to a single mode. We emphasise that the precise mode assignment in a given state may vary across training seeds, as it depends on exploration order; the robust and reproducible features are the unconditional proactive policy for Class A and the emergence of distinct, lower-overhead policies for non-urgent traffic.

Fig. 9 presents the learned Q-values for all 18 states. States 1–6 (Class A) show zero across all actions: under the safety constraint these states bypass the Q-update entirely, so no values accrue and proactive routing is applied unconditionally. States 7–18 (Class B and C) show non-zero, differentiated values across the proactive, reactive, and hybrid actions, demonstrating that

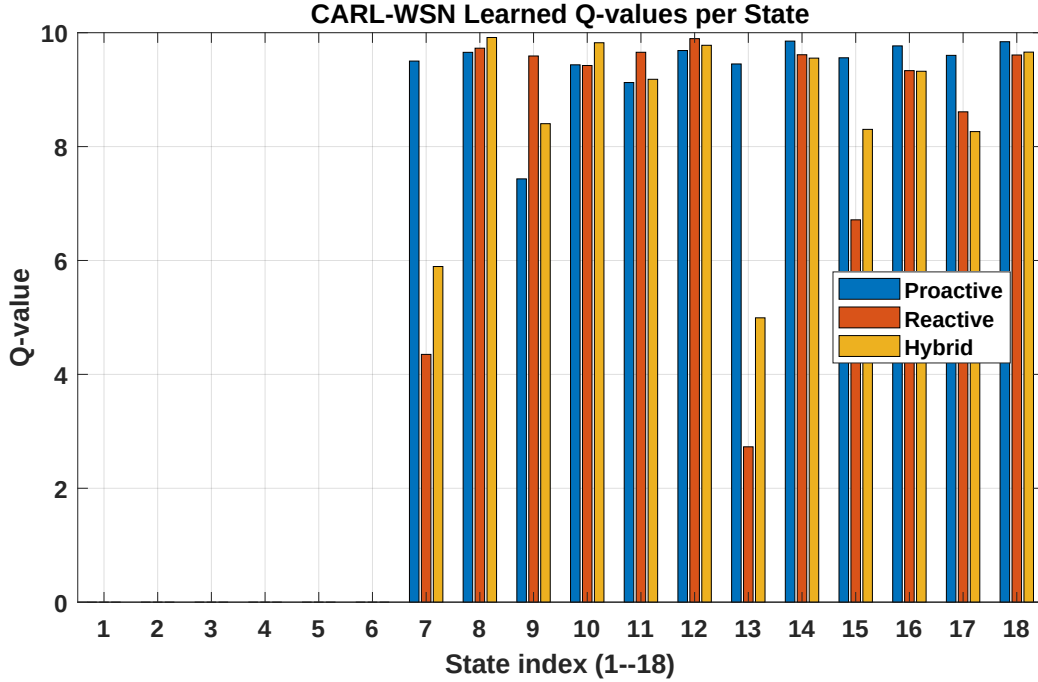


Figure 9: CARL-WSN learned Q-values for all 18 states. States 1–6 (Class A) are zero because the safety constraint excludes them from learning and forces proactive routing. States 7–18 (Class B/C) show differentiated values across the three routing modes, confirming context-dependent learning for non-emergency traffic.

823 the agent has learned context-specific action preferences for the traffic classes
824 it is permitted to optimise.

825 6.9. CARL-WSN vs RL Baselines

826 The three RL-based protocols occupy distinct regions of the design
827 space. RLCR and FQ-UCR apply reinforcement learning *within* the cluster-
828 ing paradigm — learning cluster-head selection and inter-cluster routing —
829 whereas CARL-WSN applies it to per-packet paradigm selection on a flat
830 topology. This produces a clear division of strengths rather than dominance
831 by any one protocol.

832 On network lifetime, CARL-WSN leads on the means: its FND (418)
833 exceeds RLCR’s (390) and greatly exceeds FQ-UCR’s (123), and its HND
834 (2198) is far above both clustering protocols (RLCR 1305, FQ-UCR 1172),
835 which degrade rapidly once cluster heads begin to fail. The HND gap over both
836 clustering protocols is statistically significant, as is the FND gap over FQ-UCR;

the FND difference from RLCR, however, is within seed variation and not significant. On Class A latency, CARL-WSN (24.2 ms) is significantly lower than both RLCR (34.0 ms) and FQ-UCR (35.2 ms), since neither clustering protocol prioritises emergency traffic. On routing overhead, the ordering reverses: the clustering protocols’ amortised control traffic (0.04–0.18 per packet) is below CARL-WSN’s 0.53, an inherent benefit of their steady-state cluster structure. On PDR, the three are close (CARL-WSN 90.2%, RLCR 94.1%, FQ-UCR 91.5%).

Table 6 summarises these pairwise differences. The picture is not one of Pareto dominance in either direction: CARL-WSN trades higher routing overhead for substantially better lifetime and emergency latency, while the clustering protocols make the opposite trade. Which is preferable depends on the deployment — CARL-WSN suits latency- and lifetime-critical heterogeneous traffic, whereas the clustering protocols suit overhead-constrained homogeneous monitoring.

6.10. Sensitivity Analysis

To assess robustness, we vary CARL-WSN’s principal design parameters one at a time around their default values, holding all else fixed, and re-run the mixed-traffic scenario across the full set of 20 seeds. Two groups of parameters are examined: the deferral policy (T_{defer} and $\max_{\text{defer}}$), which governs the lifetime–delivery trade, and the model parameters that lack an exact value in the literature (the relay weights, the channel shape parameters, and the link-stability threshold).

6.10.1. Deferral Policy: the Lifetime–Delivery Frontier

The deferral parameters expose a direct trade between packet delivery and network lifetime. Table 7 reports PDR and FND as T_{defer} and δ_{max} are varied. Disabling deferral ($T_{\text{defer}} = 0$) yields the highest PDR (95.1%) at the lowest FND (381), while aggressive deferral ($T_{\text{defer}} = 0.20$, $\delta_{\text{max}} = 10$) yields the highest FND (635) at the lowest PDR (66.4%). The relationship is monotonic: deferring more non-urgent traffic at near-depleted nodes extends lifetime at a proportional cost in delivery.

The operating point used throughout this paper, $T_{\text{defer}} = 0.05$ and $\max_{\text{defer}} = 3$, is selected from this frontier as the point that retains a clear lifetime advantage over all baselines (FND 418, the highest of any protocol) while keeping the delivery cost modest (PDR 90.2%, within roughly five points of the baseline group). More aggressive settings purchase additional lifetime

873 at a delivery cost we judge too steep for the marginal gain; for example,
874 raising T_{defer} to 0.15 extends FND by a further 107 rounds but reduces PDR
875 by almost 14 points. Deployments with different priorities can move along
876 this frontier by adjusting the two parameters.

877 6.10.2. Robustness to Model Parameters

878 Table 8 summarises the effect of the remaining parameters, reported as
879 the percentage swing of each metric across the swept range. CARL-WSN is
880 insensitive to the relay weight split (w_e/w_p): varying it from 0.4/0.6 to 0.8/0.2
881 changes PDR by under 1% and FND not at all, with only a mild effect on
882 HND. The link-stability threshold T_L has no measurable effect, because the
883 PRR-weighted relay metric keeps the large majority of links stable: across all
884 forwarding decisions, the link-stability score falls below the 0.7 threshold in
885 only 5.6% of cases, so varying the threshold within the 0.6–0.8 range alters
886 few decisions. The channel sharpness parameter α likewise has negligible
887 effect on PDR.

888 The one parameter with substantial effect is the PRR midpoint β , which
889 sets the distance at which links begin to fail. Increasing β from 80 to
890 100 m improves PDR, FND, and HND monotonically, as expected: a more
891 favourable channel reaches farther and delivers more. Crucially, because all
892 eight protocols share the same channel model, β shifts every protocol together
893 and does not alter CARL-WSN’s relative standing; it is a property of the
894 deployment environment rather than of the protocol.

895 In summary, CARL-WSN’s behaviour is robust to its tunable design
896 choices: delivery is essentially unchanged across the relay weights, the link-
897 stability threshold, and the channel sharpness; lifetime shows only mild
898 sensitivity to the channel sharpness ($\sim 15\%$ FND swing) and none to the other
899 two; and all metrics vary with the channel midpoint only as the underlying
900 physics dictates, equally for all protocols. The deferral policy is the one
901 parameter with a deliberate, well-characterised effect, and the chosen operating
902 point is justified by the frontier in Table 7.

903 6.11. Summary of Findings

904 Table 9 summarises CARL-WSN’s standing against the baseline group
905 across the principal metrics under mixed traffic. The results support a focused
906 set of claims. CARL-WSN delivers the **longest pre-failure lifetime** (highest
907 FND, second-highest HND), the **lowest emergency latency among the**
908 **reactive and learning-based protocols** (24.2 ms, 25.6% below AODV), and

909 **routing overhead far below the flooding- and table-based protocols**
 910 (0.53 versus 2.31–14.82). These gains are obtained at a modest, bounded
 911 cost in packet delivery (90.2% PDR, ≈ 5 points below the baselines), while
 912 energy consumption is statistically comparable to every baseline. Energy
 913 balance shows no significant difference involving CARL-WSN, and throughput
 914 is statistically comparable to the classical protocols and EH-Routing, above
 915 the two clustering protocols, and below MSLBA.

916 These findings position CARL-WSN not as a universally dominant protocol
 917 but as one that makes a specific, defensible trade: it extends network lifetime
 918 and protects emergency-traffic latency, with low control overhead relative
 919 to non-clustering protocols, in exchange for a small reduction in best-effort
 920 delivery — a trade well suited to heterogeneous, latency- and lifetime-critical
 921 IoT-WSN deployments.

922 7. Discussion

923 7.1. Lifetime Extension Without Clustering

924 The central finding of this study is that paradigm switching offers a
 925 different operating point from paradigm optimisation, rather than dominating
 926 it. CARL-WSN — which uses Q-Learning to switch *between* routing paradigms
 927 on a flat topology — achieves the longest pre-failure lifetime (highest FND,
 928 second-highest HND) and the lowest emergency latency among the learning-
 929 based protocols, while RLCR and FQ-UCR — which apply Q-Learning
 930 *within* the clustering paradigm — achieve lower steady-state control overhead.
 931 Neither approach dominates the other across all metrics; each makes a coherent
 932 trade.

933 This contradicts the intuition that clustering is uniformly superior for
 934 lifetime. Clustering concentrates forwarding at a small set of cluster heads,
 935 which must then be rotated to avoid premature death; under a realistic lossy
 936 channel this concentration produces the rapid post-FND decay visible in
 937 RLCR and FQ-UCR (HND of 1305 and 1172, the two lowest of all protocols).
 938 CARL-WSN instead distributes forwarding across many energy-weighted
 939 relay paths and defers traffic only at near-depleted nodes, which delays both
 940 the first death and the progression to half-death. The cost of this flat-routing
 941 approach is higher per-packet control overhead than clustering’s amortised
 942 steady state — but that overhead remains far below the flooding- and table-
 943 based protocols, and it buys a lifetime advantage that clustering does not
 944 provide.

945 The broader implication for RL-based WSN routing is therefore not that
 946 paradigm switching is universally more efficient, but that the choice of *what*
 947 the agent controls — the routing paradigm itself, versus decisions within a
 948 fixed paradigm — determines which region of the cost–lifetime trade space
 949 the protocol occupies. CARL-WSN’s contribution is to show that learned
 950 per-packet paradigm selection on a flat topology can extend lifetime beyond
 951 clustering while keeping overhead well below classical flat protocols.

952 7.2. *Energy-Aware Relay Selection and Network Lifetime*

953 CARL-WSN’s energy-aware relay selection in (6) addresses a well-known
 954 vulnerability of flat-routing protocols: nodes near the base station forward
 955 disproportionate traffic and deplete early. The classical flat protocols (AODV,
 956 DSDV, ZRP) share an identical FND of 327 under mixed traffic, since they
 957 select next hops on geographic progress alone without regard to residual
 958 energy. It is therefore the relay-selection and deferral mechanisms, rather
 959 than the choice of routing paradigm itself, that account for CARL-WSN’s
 960 lifetime advantage; this is confirmed directly by Table 7, where disabling
 961 deferral lowers FND to 381 — on par with EH-Routing (380) and below RLCR
 962 (390) — so that the lead over the lifetime-oriented baselines is produced by
 963 the deferral policy that the urgency signal makes possible, not by paradigm
 964 switching.

965 By weighting relay selection toward residual energy as well as progress,
 966 and by bounding deferral at near-depleted nodes, CARL-WSN raises FND
 967 to 418 — the highest of all eight protocols, ahead of the energy-harvesting
 968 (EH-Routing, 380) and clustering (RLCR, 390) baselines, and 28% above the
 969 classical flat protocols. On half-node-death, CARL-WSN reaches HND = 2198,
 970 second only to MSLBA (2340) and well above the classical protocols (1877)
 971 and both clustering protocols (1305 and 1172). The combination of energy-
 972 aware relaying and bounded deferral thus delays both the onset and the
 973 progression of network failure, giving a flat-routing protocol a lifetime profile
 974 competitive with, and on FND exceeding, the dedicated lifetime-oriented
 975 baselines — while retaining the low control overhead characteristic of flat
 976 routing relative to flooding and table maintenance.

977 It is important to state the cost of this lifetime gain plainly. The bounded
 978 deferral mechanism withholds a fraction of non-urgent Class B/C packets at
 979 near-depleted nodes, which reduces packet delivery to 90.2% under mixed
 980 traffic, roughly five points below the baseline group. The lifetime advantage
 981 and the delivery reduction are two sides of the same mechanism; the operating

point adopted here was chosen (Section 6.10) to keep the delivery cost modest while preserving the lifetime gain.

7.3. Safety Constraints in RL-Based Routing

A practical concern with deploying RL agents in safety-critical systems is the risk that the learned policy may violate hard constraints during exploration or after convergence to a suboptimal policy. CARL-WSN addresses this through the hard safety constraint in (20): Class A emergency packets are routed via proactive mode unconditionally, bypassing the ε -greedy selection entirely. This constraint is not a limitation of the RL agent but a deliberate architectural decision that separates safety-critical routing (rule-based) from efficiency-optimised routing (learned).

Neither RLCR [8] nor FQ-UCR [9] implements such a constraint because neither protocol differentiates traffic by urgency class. In a deployment where both emergency and routine traffic coexist, the absence of safety constraints means that the RL agent’s exploration phase could route emergency packets via suboptimal paths with unacceptable latency. CARL-WSN eliminates this risk by design.

7.4. Q-Learning Convergence Behaviour

The convergence analysis in Fig. 7 shows that the Q-table stabilises within approximately 1150 rounds, with maximum update magnitudes decreasing from 10^0 to 10^{-3} . The relatively fast convergence is attributable to the compact state space (18 states): with only 54 Q-values to learn, the agent visits each state-action pair frequently enough to achieve convergence early in the network’s lifetime, well before half-node death and while most nodes remain alive.

The epsilon decay schedule ($\varepsilon_0 = 0.1$, $\delta = 0.998$, $\varepsilon_{\min} = 0.01$) balances exploration and exploitation effectively. After 1150 rounds, ε has reached its floor $\varepsilon_{\min} = 0.01$, meaning the agent selects the learned best action approximately 99% of the time while still occasionally exploring alternatives. This residual exploration allows the agent to adapt if network conditions change — for example, if relay nodes die and previously optimal paths become unavailable. The learned policy in Fig. 8 is contextually meaningful, selecting proactive routing for emergency traffic, reactive routing for non-urgent traffic on moderate-to-healthy nodes, and hybrid routing as an intermediate choice for moderate-energy telemetry nodes. These preferences emerge entirely from the reward signal without any manually programmed rules — the agent

1018 independently discovers that different traffic classes and network conditions
1019 warrant different routing paradigms.

1020 *7.5. Context-Awareness as a Unique Capability*

1021 Table 5 demonstrates that CARL-WSN is the only protocol whose overhead
1022 varies with traffic class composition. This context-awareness has practical
1023 implications beyond overhead reduction. In a real smart city deployment,
1024 traffic composition changes dynamically: a fire event increases Class A traffic
1025 temporarily, while normal conditions are dominated by Class B telemetry.
1026 CARL-WSN automatically adjusts its routing behaviour to match the current
1027 traffic mix — using more proactive routing during emergencies and switching
1028 to energy-efficient reactive routing during normal operations. This adaptive
1029 overhead management is a direct consequence of the class-dependent reward
1030 function and cannot be replicated by protocols that treat all traffic identically.

1031 *7.6. Practical Deployment Considerations*

1032 CARL-WSN requires each node to maintain a Q-table of 216 bytes, an
1033 ACK history buffer of 80 bytes, and state computation variables of 12 bytes —
1034 a total memory overhead of 308 bytes beyond the standard routing table. This
1035 is deployable on widely available sensor platforms including TelosB (10 KB
1036 RAM), MicaZ (4 KB RAM), and Arduino-class devices without modification.

1037 The Q-Learning agent operates in $O(1)$ time per packet, adding under
1038 0.1 ms to the per-packet processing time. In a network generating one packet
1039 per node per round, the total agent computation across 100 nodes is under
1040 10 ms per round — negligible compared to the radio transmission time.

1041 Regarding scalability, CARL-WSN’s context classifier operates in $O(1)$
1042 time regardless of network size N , as the state is computed from local
1043 information only (the node’s own energy, its own link stability, and the
1044 packet’s class tag). The proactive routing table scales $O(N)$ in memory,
1045 which is standard for all proactive protocols and acceptable for the node
1046 densities typical of smart city WSN deployments. The Q-table size ($18 \times$
1047 3) is independent of N — adding more nodes does not increase the state
1048 space. CARL-WSN’s overhead advantage over the flooding- and table-based
1049 protocols is expected to persist or grow with larger N : AODV’s flood cost
1050 scales with network size, whereas CARL-WSN’s per-packet control depends
1051 only on the local neighbour count. Relative to the clustering protocols, whose
1052 amortised overhead is lower than CARL-WSN’s, the comparison at larger

1053 N is an open question that the present single-size study cannot settle, and
1054 which we identify as future work.

1055 7.7. Limitations

1056 This study has five limitations that should be acknowledged. First, while
1057 all protocols are evaluated on a common distance-dependent lossy-channel
1058 model with per-hop acknowledgement and retransmission — a substantial
1059 step beyond idealised delivery — this model does not capture multipath
1060 fading, shadowing, temporal correlation, or interference from co-deployed
1061 networks. Second, node mobility is not modelled — all nodes are static after
1062 deployment; mobile WSN topologies where link stability varies dynamically
1063 due to node movement represent a natural extension. Third, the latency
1064 values are simulated rather than measured on physical hardware. Fourth,
1065 all results are reported for a single network size ($N = 100$); scalability to
1066 larger deployments ($N = 500+$) is left for future validation. Fifth, the Q-
1067 Learning agent uses a fixed reward weight vector per traffic class; an adaptive
1068 weight-tuning mechanism that adjusts weights based on network congestion
1069 or remaining energy budget could further improve performance.

1070 Future work should validate CARL-WSN on a real sensor testbed (e.g.,
1071 Raspberry Pi or Arduino with IEEE 802.15.4 radios) to confirm that the
1072 simulated performance advantages hold under realistic channel conditions
1073 and hardware constraints.

1074 8. Conclusion

1075 This paper presented CARL-WSN, a Context-Aware Reinforcement Learn-
1076 ing protocol for hybrid routing in heterogeneous IoT-driven Wireless Sensor
1077 Networks. CARL-WSN formulates routing paradigm selection as a Markov
1078 Decision Process and uses a tabular Q-Learning agent to dynamically select
1079 between proactive, reactive, and hybrid routing on a per-packet basis. The
1080 agent observes a three-dimensional context vector comprising application
1081 urgency class, normalised residual energy, and link stability score, and learns
1082 class-dependent routing policies through a multi-objective reward function
1083 that weights latency, overhead, and energy preservation according to traffic
1084 priority. A hard safety constraint guarantees sub-100 ms delivery for emer-
1085 gency traffic regardless of the learned policy, and energy-aware relay selection
1086 combined with bounded critical-node packet deferral extends network lifetime
1087 beyond all evaluated baselines.

1088 A distinguishing feature of the evaluation is that all eight protocols are
 1089 re-implemented on a common distance-dependent lossy-channel model with
 1090 per-hop acknowledgement and bounded retransmission, so that packet delivery,
 1091 latency, and energy consumption emerge from identical physical assumptions
 1092 rather than from protocol-specific idealisations. Across four traffic scenarios
 1093 and 20 independent seeds (640 runs), CARL-WSN’s behaviour under this
 1094 unified methodology can be summarised as follows:

- 1095 • **Longest pre-failure lifetime.** CARL-WSN attains the highest mean
 1096 first-node-death round of all eight protocols ($FND = 418$), 28% above
 1097 the classical flat protocols, and the second-highest mean half-node-
 1098 death round ($HND = 2198$), behind only MSLBA. Tukey HSD analysis
 1099 confirms these lifetime gains are statistically significant over the classical
 1100 protocols and FQ-UCR (and, for HND, over RLCR), while CARL-WSN
 1101 remains statistically comparable to the energy-oriented baselines EH-
 1102 Routing and MSLBA. This lifetime advantage is produced by the energy-
 1103 aware relay selection and bounded deferral that the per-packet urgency
 1104 signal enables, rather than by paradigm switching itself: as Table 7
 1105 shows, disabling deferral returns FND to the level of the energy-oriented
 1106 baselines.
- 1107 • **Low emergency latency.** CARL-WSN delivers Class A traffic in
 1108 24.2 ms, a 25.6% reduction versus AODV and among the lowest of all
 1109 protocols, with the safety constraint guaranteeing sub-100 ms delivery
 1110 unconditionally.
- 1111 • **Low overhead relative to flooding and table maintenance.**
 1112 CARL-WSN incurs 0.53 control packets per data packet, far below
 1113 AODV (14.82), ZRP (5.91), and DSDV (2.31). It does not match the
 1114 clustering protocols RLCR and FQ-UCR, whose control traffic is amor-
 1115 tised over long re-clustering intervals (0.04–0.18 per packet); this is an
 1116 inherent benefit of the clustering paradigm that CARL-WSN trades for
 1117 per-packet adaptation and freedom from cluster-head energy hotspots.
- 1118 • **Context-adaptive overhead.** CARL-WSN is the only evaluated
 1119 protocol whose overhead varies with traffic-class composition — from
 1120 1.00 for pure emergency traffic to 0.32 for pure non-urgent traffic — em-
 1121 pirically confirming that the agent learns class-specific routing policies.

1122 • **A modest, bounded delivery trade.** These gains are obtained
 1123 at a packet-delivery ratio of 90.2% under mixed traffic, roughly five
 1124 points below the baseline group, arising from the deferral of non-urgent
 1125 traffic at near-depleted nodes. On the neutral metrics, CARL-WSN is
 1126 statistically comparable to all baselines in energy consumption, shows
 1127 no significant energy-balance difference, and matches the classical and
 1128 energy-harvesting protocols in throughput (below MSLBA, above the
 1129 clustering protocols).

1130 The Q-table converges within approximately 1150 rounds, occupies only
 1131 216 bytes of RAM in single precision, and the agent operates in $O(1)$ time
 1132 per packet, making CARL-WSN deployable on resource-constrained sensor
 1133 hardware without modification. The learned policy autonomously discovers
 1134 that emergency traffic is best served by proactive routing, non-urgent traffic
 1135 by reactive routing, and moderate-energy telemetry nodes by an intermediate
 1136 hybrid strategy — all without manually programmed rules.

1137 The central finding of this work is that the choice of *what* a reinforcement-
 1138 learning agent controls — the routing paradigm itself, rather than decisions
 1139 within a fixed paradigm — determines which region of the cost–lifetime trade
 1140 space a protocol occupies. By learning per-packet paradigm selection on
 1141 a flat topology, CARL-WSN extends network lifetime beyond clustering-
 1142 based protocols and protects emergency-traffic latency, while keeping control
 1143 overhead well below classical flooding- and table-based protocols, at the cost
 1144 of a small reduction in best-effort delivery. This positions paradigm switching
 1145 not as a universally dominant strategy but as a distinct and well-characterised
 1146 operating point for heterogeneous, latency- and lifetime-critical IoT-WSN
 1147 deployments.

1148 Four directions for future work are identified. First, validating CARL-WSN
 1149 on a physical sensor testbed to confirm that the simulated advantages hold
 1150 under real channel and hardware conditions. Second, extending the protocol
 1151 to mobile WSN topologies where link stability varies dynamically due to node
 1152 movement. Third, scaling the evaluation to larger networks ($N = 500+$),
 1153 where the overhead comparison against clustering protocols in particular
 1154 remains an open question. Fourth, investigating adaptive reward-weight
 1155 tuning that adjusts the class-dependent weight vectors based on real-time
 1156 network congestion and remaining energy budget.

References

- [1] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, E. Cayirci, Wireless sensor networks: a survey, *Computer Networks* 38 (4) (2002) 393–422. doi:10.1016/S1389-1286(01)00302-4.
- [2] N. Zohrabi, et al., A review of IoT-based smart city development and management, *Smart Cities* 7 (3) (2024) 61. doi:10.3390/smartcities7030061.
- [3] C. E. Perkins, P. Bhagwat, Highly dynamic destination-sequenced distance-vector routing for mobile computers, in: *Proc. ACM SIGCOMM*, 1994, pp. 234–244.
- [4] S. R. Das, C. E. Perkins, E. M. Belding-Royer, Ad hoc on-demand distance vector (AODV) routing, Request for Comments RFC 3561, Internet Engineering Task Force (2003). doi:10.17487/RFC3561. URL <https://datatracker.ietf.org/doc/rfc3561>
- [5] D. B. Johnson, D. A. Maltz, Dynamic source routing in ad hoc wireless networks, in: T. Imielinski, H. F. Korth (Eds.), *Mobile Computing*, Vol. 353 of The Kluwer International Series in Engineering and Computer Science, Kluwer Academic Publishers, Boston, MA, 1996, pp. 153–181. doi:10.1007/978-0-585-29603-6_5.
- [6] I. M. Selim, N. S. Abdelrehem, W. M. Alayed, H. M. Elbadawy, R. A. Sadek, MANET routing protocols’ performance assessment under dynamic network conditions, *Applied Sciences* 15 (6) (2025) 2891. doi:10.3390/app15062891. URL <https://www.mdpi.com/2076-3417/15/6/2891>
- [7] Z. J. Haas, M. R. Pearlman, The performance of query control schemes for the zone routing protocol, *IEEE/ACM Trans. Networking* 9 (4) (2001) 427–438.
- [8] I. Daanoune, A. Baghdad, Intelligent energy-aware routing in IoT-based WSNs by reinforcement learning, *Ad Hoc Networks* 188 (2026) 104244. doi:10.1016/j.adhoc.2026.104244. URL <https://www.sciencedirect.com/science/article/pii/S1570870526001101>

- 1189 [9] Z. Wang, J. Duan, An unequal clustering and multi-hop routing protocol
1190 based on fuzzy logic and Q-learning in WSNs, *Entropy* 27 (2) (2025) 118.
1191 doi:10.3390/e27020118.
1192 URL <https://www.mdpi.com/1099-4300/27/2/118>
- 1193 [10] A. M. Khedr, O. Alfawaz, P. V. Pravija Raj, W. Osamy, Advancing
1194 IoT-driven WSNs with context-aware routing: A comprehensive review,
1195 *Computer Science Review* 58 (2025) 100803.
- 1196 [11] K. Akkaya, M. Younis, A survey on routing protocols for wireless sensor
1197 networks, *Ad Hoc Networks* 3 (3) (2005) 325–349. doi:10.1016/j.
1198 *adhoc*.2003.09.010.
- 1199 [12] T. H. Clausen, P. Jacquet, Optimized link state routing protocol (OLSR),
1200 Request for Comments RFC 3626, Internet Engineering Task Force (2003).
1201 doi:10.17487/RFC3626.
1202 URL <https://datatracker.ietf.org/doc/rfc3626>
- 1203 [13] Nurul Hidayah Ahmad Zukri, Ahmad Suki Che Mohamed Arif, Mo-
1204 hammed Alsamman, Comparative analysis of reactive and proactive
1205 mobility management in named data networking, *IAENG Int. J. Com-*
1206 *puter Science* 52 (2) (2025).
- 1207 [14] M. Karthikeyan, D. Manimegalai, K. Rajagopal, Optimizing energy
1208 efficiency in wireless sensor networks: dynamic routing with capuchin
1209 search algorithm, *Multimedia Tools and Applications* 84 (15) (2025)
1210 15453–15477. doi:10.1007/s11042-024-19625-7.
1211 URL <https://doi.org/10.1007/s11042-024-19625-7>
- 1212 [15] S. K. Chaurasiya, S. Mondal, A. Biswas, A. Nayyar, M. A. Shah, R. Baner-
1213 jee, An energy-efficient hybrid clustering technique (EEHCT) for IoT-
1214 based multilevel heterogeneous wireless sensor networks, *IEEE Access* 11
1215 (2023) 25941–25958. doi:10.1109/ACCESS.2023.3254594.
1216 URL <https://ieeexplore.ieee.org/document/10064298/>
- 1217 [16] C. Lei, An energy-aware cluster-based routing in the internet of things
1218 using particle swarm optimization algorithm and fuzzy clustering, *J.*
1219 *Engineering and Applied Science* 71 (1) (2024) 135. doi:10.1186/
1220 *s44147-024-00464-0*.
1221 URL <https://doi.org/10.1186/s44147-024-00464-0>

- 1222 [17] P. Nadanam, N. Kumaratharan, M. Anousouya Devi, Innovative hybrid
1223 framework for routing and clustering in wireless sensor networks with
1224 quantum optimization, *Cybernetics and Systems* 57 (3) (2026) 449–484.
1225 doi:10.1080/01969722.2025.2521710.
1226 URL <https://doi.org/10.1080/01969722.2025.2521710>
- 1227 [18] C. Perera, A. Zaslavsky, P. Christen, D. Georgakopoulos, Context aware
1228 computing for the internet of things: A survey, *IEEE Communications*
1229 *Surveys & Tutorials* 16 (1) (2014) 414–454. doi:10.1109/SURV.2013.
1230 042313.00197.
1231 URL <https://ieeexplore.ieee.org/abstract/document/6512846>
- 1232 [19] A. Monteiro, E. Souto, R. Pazzi, M. Nogueira, Context-aware network
1233 selection in heterogeneous wireless networks, *Computer Communications*
1234 135 (2019) 1–15. doi:10.1016/j.comcom.2018.11.006.
1235 URL [https://linkinghub.elsevier.com/retrieve/pii/
1236 S014036641730155X](https://linkinghub.elsevier.com/retrieve/pii/S014036641730155X)
- 1237 [20] M. M. Sithik, B. M. Kumar, Intelligent agent based virtual clus-
1238 tering and multi-context aware routing for congestion mitigation in
1239 secure RPL-IoT environment, *Ad Hoc Networks* 137 (2022) 102972.
1240 doi:10.1016/j.adhoc.2022.102972.
1241 URL [https://www.sciencedirect.com/science/article/pii/
1242 S1570870522001470](https://www.sciencedirect.com/science/article/pii/S1570870522001470)
- 1243 [21] V. Karagiannis, P. A. Frangoudis, S. Dustdar, S. Schulte, Context-aware
1244 routing in fog computing systems, *IEEE Trans. Cloud Computing* 11 (1)
1245 (2023) 532–549. doi:10.1109/TCC.2021.3102996.
1246 URL <https://ieeexplore.ieee.org/abstract/document/9508866>
- 1247 [22] A. A. Hasan, X. Fang, S. Latif, A. Iqbal, Context-aware trust prediction
1248 for optimal routing in opportunistic IoT systems, *Sensors* 25 (12) (2025)
1249 3672. doi:10.3390/s25123672.
1250 URL <https://www.mdpi.com/1424-8220/25/12/3672>
- 1251 [23] G. A. Akpakwu, G. P. Hancke, A. M. Abu-Mahfouz, CACC: Context-
1252 aware congestion control approach for lightweight CoAP/UDP-based
1253 internet of things traffic, *Transactions on Emerging Telecommunications*
1254 *Technologies* 31 (2) (2020) e3822. doi:10.1002/ett.3822.
1255 URL <https://onlinelibrary.wiley.com/doi/10.1002/ett.3822>

- [24] D. Karunkuzhali, B. Meenakshi, K. Lingam, OQR-SC: An optimal QoS aware routing technique for smart cities using IoT enabled wireless sensor networks, *Wireless Personal Communications* 125 (4) (2022) 3575–3602. doi:10.1007/s11277-022-09725-8. URL <https://link.springer.com/10.1007/s11277-022-09725-8>
- [25] D. Karunkuzhali, B. Meenakshi, K. Lingam, A QoS-aware routing approach for internet of things-enabled wireless sensor networks in smart cities, *Multimedia Tools and Applications* 84 (17) (2024) 17951–17977. doi:10.1007/s11042-024-18125-y. URL <https://link.springer.com/10.1007/s11042-024-18125-y>
- [26] F. H. El-Fouly, M. Kachout, Y. Alharbi, J. S. Alshudukhi, A. Alanazi, R. A. Ramadan, Environment-aware energy efficient and reliable routing in real-time multi-sink wireless sensor networks for smart cities applications, *Applied Sciences* 13 (1) (2023) 605. doi:10.3390/app13010605. URL <https://www.mdpi.com/2076-3417/13/1/605>
- [27] S. A. Obaid, A. J. Salman, Load-balancing in multi-sink IoT networks, *Int. J. Scientific and Engineering Research (IJSER)* 2 (2) (2023) 58–64. doi:10.58564/IJSER.2.2.2023.71. URL <https://ijser.aliraqia.edu.iq/index.php/ijser/article/view/71>
- [28] S. Hanif, A. M. Khedr, Z. Al Aghbari, D. P. Agrawal, Opportunistically exploiting internet of things for wireless sensor network routing in smart cities, *Journal of Sensor and Actuator Networks* 7 (4) (2018) 46. doi:10.3390/jsan7040046. URL <https://www.mdpi.com/2224-2708/7/4/46>
- [29] E. Inga, J. Inga, A. Ortega, Novel approach sizing and routing of wireless sensor networks for applications in smart cities, *Sensors* 21 (14) (2021) 4692. doi:10.3390/s21144692. URL <https://www.mdpi.com/1424-8220/21/14/4692>
- [30] P. Ekler, J. Levendovszky, D. Pasztor, Energy aware IoT routing algorithms in smart city environment, *IEEE Access* 10 (2022) 87733–87744. doi:10.1109/ACCESS.2022.3199757. URL <https://ieeexplore.ieee.org/document/9861603/>

- 1289 [31] R. Priyadarshi, Energy-efficient routing in wireless sensor networks: A
1290 meta-heuristic and artificial intelligence-based approach: A comprehensive review, Archives of Computational Methods in Engineering 31 (4)
1291 (2024) 2109–2137. doi:10.1007/s11831-023-10039-6.
1292 URL <https://doi.org/10.1007/s11831-023-10039-6>
1293
- 1294 [32] R. Priyadarshi, R. R. Kumar, R. Ranjan, P. V. Kumar, AI-based routing
1295 algorithms improve energy efficiency, latency, and data reliability in
1296 wireless sensor networks, Scientific Reports 15 (1) (2025) 22292. doi:
1297 10.1038/s41598-025-08677-w.
1298 URL <https://www.nature.com/articles/s41598-025-08677-w>
- 1299 [33] M. Y. Yilmaz, B. Ustuner, I. M. Gul, H. A. Cirpan, Sustainable commu-
1300 nication in 5g/6g wireless sensor networks: A survey on energy-efficient
1301 collaborative routing, ITU Journal of Wireless Communications and
1302 Cybersecurity 2 (1) (2025) 11–26.
1303 URL [https://dergipark.org.tr/en/pub/itujwcc/article/](https://dergipark.org.tr/en/pub/itujwcc/article/1639314)
1304 1639314
- 1305 [34] S. Sudha, A. Srinivasan, T. G. Devi, G. Rajkumar, Intelligent routing for
1306 5g-enabled wireless sensor networks using hybrid learning approaches, in:
1307 2025 IEEE 17th International Conference on Computational Intelligence
1308 and Communication Networks (CICN), 2025, pp. 141–145, ISSN: 2472-
1309 7555. doi:10.1109/CICN67655.2025.11368058.
1310 URL <https://ieeexplore.ieee.org/document/11368058>
- 1311 [35] N. R. Saadallah, S. A. Alabady, A comprehensive study on energy-
1312 efficient-based routing protocols in the internet of things part i: definition
1313 and classification, Iran Journal of Computer Science 7 (3) (2024) 631–661.
1314 doi:10.1007/s42044-024-00180-1.
1315 URL <https://doi.org/10.1007/s42044-024-00180-1>
- 1316 [36] P. Kumar, A. Sharma, R. Grover, N. Kumar, V. Kuchhal, S. Singh, Ex-
1317 ploring energy-efficient routing in IoT-based WSNs: A WoS bibliometric-
1318 based review, in: 2025 8th International Conference on Comput-
1319 ing Methodologies and Communication (ICCMC), 2025, pp. 401–406.
1320 doi:10.1109/ICCMC65190.2025.11140596.
1321 URL <https://ieeexplore.ieee.org/abstract/document/11140596>

- [37] A. Chu, T. Zhang, C. Wang, LEACH protocol evolution in WSN: A review of energy consumption optimization and security reinforcement, *Sensors* 26 (7) (2026) 2272. doi:10.3390/s26072272. URL <https://www.mdpi.com/1424-8220/26/7/2272>
- [38] P. Gupta, S. Tripathi, S. Singh, Energy-efficient routing protocols for cluster-based heterogeneous wireless sensor network (Het-WSN)—strategies and challenges: A review, in: A. Khanna, D. Gupta, Z. Pólkowski, S. Bhattacharyya, O. Castillo (Eds.), *Data Analytics and Management*, Vol. 54, Springer Singapore, 2021, pp. 853–878, series Title: *Lecture Notes on Data Engineering and Communications Technologies*. doi:10.1007/978-981-15-8335-3_65. URL http://link.springer.com/10.1007/978-981-15-8335-3_65
- [39] C. M. Prashanth, et al., Cluster based energy efficient routing protocol for heterogeneous wireless sensor networks, *Concurrency and Computation: Practice and Experience* 35 (7) (2023) e7693. doi:10.1002/cpe.7693.
- [40] M. M. Saleh, R. S. Abdulrahman, A. J. Salman, Energy-harvesting and energy-aware routing algorithm for heterogeneous energy WSNs, *Indonesian J. Electrical Engineering and Computer Science (IJECS)* 24 (2) (2021) 910–920.
- [41] W. Heinzelman, A. Chandrakasan, H. Balakrishnan, Energy-efficient communication protocol for wireless microsensor networks, in: *Proc. 33rd Annual Hawaii Int. Conf. System Sciences (HICSS)*, IEEE, 2000, pp. 1–10.
- [42] M. Zuniga, B. Krishnamachari, Analyzing the transitional region in low power wireless links, in: *2004 First Annual IEEE Communications Society Conference on Sensor and Ad Hoc Communications and Networks (SECON)*, 2004, pp. 517–526. doi:10.1109/SAHCN.2004.1381954.
- [43] IEEE, IEEE standard for low-rate wireless networks, *IEEE Std 802.15.4-2020* (2020). doi:10.1109/IEEESTD.2020.9144691.
- [44] A. Woo, D. Culler, Evaluation of efficient link reliability estimators for low-power wireless networks, *Tech. Rep. UCB/CSD-03-1270*, EECS Dept., Univ. of California, Berkeley (2003).

1354 [45] D. Bhatt, et al., Implementation of novel learning based energy ef-
1355 ficient routing protocols in wireless sensor networks for Internet of
1356 Things use cases, Discover Computing 28 (1) (2025) 1–24. doi:
1357 10.1007/s10791-025-09718-8.

Algorithm 1: CARL-WSN Per-Packet Routing with Q-Learning

Input: Packet p with class U , node i , network state $\mathcal{N}$, Q-table Q , exploration rate ε , per-node deferral counter c_{def}^i (init. 0)

Output: Updated network state, updated Q-table

```
// Step 1: Compute context vector
1  $U \leftarrow \text{READCLASSTAG}(p)$ ;
2  $E_r^i \leftarrow E_i(t)/E_0^i$ ;
3  $L_s^i \leftarrow \sum_{k=0}^{W-1} w_k \cdot a_k^i$ ;
4  $s \leftarrow \text{STATEINDEX}(U, E_r^i, L_s^i)$ ;
5 if  $U = A$  then
    // Step 2: Safety constraint -- emergency traffic
    // bypasses learning
6      $a \leftarrow \text{proactive}$ ;
7 else
    /* Step 3: Bounded critical-node deferral */
8     if  $E_r^i < T_{\text{defer}}$  and  $U \in \{B, C\}$  and  $c_{\text{def}}^i < \delta_{\text{max}}$  then
9          $c_{\text{def}}^i \leftarrow c_{\text{def}}^i + 1$ ;
10        Defer packet (minimal listening energy cost);
11        return;
12    else
13         $c_{\text{def}}^i \leftarrow 0$ ; // forced or normal transmit ends the
        // deferral run
14    end
15 end
    // Step 4:  $\varepsilon$ -greedy action selection
16 if  $\text{rand}() < \varepsilon$  then
17      $a \leftarrow \text{random action from } \mathcal{A}$ ;
18 else
19      $a \leftarrow \arg \max_{a'} Q(s, a')$ ;
20 end
    // Step 5: Route with energy-aware relay
21  $j^* \leftarrow \arg \max_{j \in \mathcal{N}_i} [(0.4 \cdot \text{progress}(j) + 0.6 \cdot E_r^j) \cdot \text{PRR}(d_{ij})]$ ;
22 Route packet via mode  $a$  using relay  $j^*$ ;
    // Step 6: Observe outcome and compute reward
23  $R \leftarrow w_1^U \cdot R_{\text{lat}} + w_2^U \cdot R_{\text{oh}} + w_3^U \cdot R_{\text{en}}$ ;
24  $s' \leftarrow \text{STATEINDEX}(U, E_r^{i, \text{new}}, L_s^{i, \text{new}})$ ;
    // Step 7: Q-table update (skip for Class A)
25 if  $U \neq A$  then
26      $Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$ ;
27 end
28  $\varepsilon \leftarrow \max(\varepsilon \cdot \delta, \varepsilon_{\text{min}})$ ;
```

Table 3: Simulation Parameters

Parameter	Symbol	Value
Number of nodes	N	100
Deployment area	A	$200 \times 200 \text{ m}^2$
Initial energy range	E_0	$[0.5, 2.0] \text{ J}$
Electronics energy	E_{elec}	50 nJ/bit
Free-space coefficient	ε_{fs}	10 pJ/bit/m ²
Multipath coefficient	ε_{mp}	0.0013 pJ/bit/m ⁴
Packet size	L	4000 bits
PRR midpoint	β_{ch}	90 m
PRR sharpness	α_{ch}	0.18
Max retransmissions	$R_{\max}^{ARQ}$	3
Communication range	R_{tx}	100 m
ACK window size	W	30
Simulation rounds	R	8000
Independent runs	—	20 (seeds 42–61)
<i>Q-Learning hyperparameters</i>		
Learning rate	α	0.1
Discount factor	γ	0.9
Initial exploration	ε_0	0.1
Exploration decay	δ	0.998
Minimum exploration	$\varepsilon_{\min}$	0.01
Deferral threshold	T_{defer}	0.05
Max consecutive deferrals	$\delta_{\max}$	3
EWMA smoothing factor	α_{ewma}	0.5
Relay progress weight	w_p	0.4
Relay energy weight	w_e	0.6

Table 4: Complete Results — Mixed Traffic Scenario (mean across 20 seeds)

Protocol	FND	HND	PDR	LatA (ms)	AvgE (J)	Gini	Overhead	Tput (pkt/rnd)
CARL-WSN	418	2198	90.2%	24.2	0.024	0.200	0.53	37.8
AODV	327	1877	94.9%	32.5	0.024	0.196	14.82	36.7
DSDV	327	1877	94.9%	24.4	0.024	0.196	2.31	36.7
ZRP	327	1877	94.9%	28.2	0.024	0.196	5.91	36.7
EH-Routing	380	2126	96.0%	24.4	0.024	0.203	1.00	38.2
MSLBA	362	2340	96.9%	21.1	0.024	0.193	1.00	48.5
RLCR	390	1305	94.1%	34.0	0.024	0.200	0.04[‡]	23.2
FQ-UCR	123	1172	91.5%	35.2	0.024	0.198	0.04[‡]	19.5

Bold = best value in column. [‡]Re-clustering-only accounting; 0.10–0.18 under TDMA accounting (see text).

[‡]For the clustering protocols (RLCR, FQ-UCR), the overhead column reports re-clustering control packets per data packet; their per-slot TDMA scheduling overhead (0.10–0.18) is accounted separately and shown in Fig. 3.

Table 5: Routing Overhead per Traffic Scenario — Context-Awareness Validation

Protocol	Class A	Class B	Class C	Mixed	Adapts?
CARL-WSN	1.00	0.32	0.32	0.53	Yes
AODV	14.82	14.82	14.82	14.82	No
EH-Routing	1.00	1.00	1.00	1.00	No
MSLBA	1.00	1.00	1.00	1.00	No

Table 6: CARL-WSN versus RL Baselines — Mixed Traffic (CARL-WSN value vs. baseline value)

Comparison	FND	HND	LatA (ms)	Overhead	PDR
CARL-WSN	418	2198	24.2	0.53	90.2%
RLCR	390	1305	34.0	0.04–0.18	94.1%
FQ-UCR	123	1172	35.2	0.04–0.10	91.5%

CARL-WSN leads on lifetime (FND, HND) and emergency latency;
the clustering baselines lead on overhead and slightly on PDR.

Table 7: Deferral Policy Sensitivity — Mixed Traffic (20 seeds)

T_{defer}	δ_{max}	PDR (%)	FND	HND
0.00 (off)	—	95.1	381	2117
0.05	3	90.2	418	2198
0.05	5	88.9	428	2220
0.10	3	85.1	455	2295
0.10	5	82.7	475	2342
0.15	5	76.5	525	2478
0.20	10	66.4	635	2787

Table 8: Model-Parameter Sensitivity — Metric Swing (%) Across Swept Range, Mixed Traffic, 20 Seeds

Parameter (range)	PDR	FND	HND	LatA	OH
w_e/w_p (0.4/0.6)	0.6	0.0	4.7	1.0	0.4
PRR β (80–100)	10.1	64.8	28.5	16.2	2.7
PRR α (0.12–0.25)	0.2	15.2	6.5	6.9	4.3
T_L (0.6–0.8)	0.0	0.0	0.0	0.0	0.0

Swing = $100 \times (\max - \min) / \text{mean}$ across the swept values.

Table 9: CARL-WSN Standing on Principal Metrics — Mixed Traffic

Metric	CARL-WSN result
Network lifetime (FND)	Highest of all eight protocols (418)
Network lifetime (HND)	Second-highest (2198), behind MSLBA
Class A latency	25.6% below AODV; among the lowest overall
Routing overhead	Far below AODV/ZRP/DSDV; above clustering
Packet delivery	90.2% (≈ 5 pts below baselines)
Energy consumption	Statistically comparable to all baselines
Energy balance (Gini)	No significant difference involving CARL-WSN
Throughput	Comparable to classical and EH-Routing; above clustering; below MSLBA